# Long Island Clustering Workflow

AMS 597 Project Team

April 07, 2026

## Phase 1: Environment and Reproducibility Setup

This file is the main reproducible workflow for the Long Island clustering analysis. It is meant to run from a clean R session and use `li_accidents_clean.csv` as the only project dataset input.

```r
required_packages <- c(
  "data.table",
  "dplyr",
  "lubridate",
  "clustMixType",
  "dbscan",
  "cluster",
  "fpc",
  "ggplot2",
  "geosphere",
  "factoextra",
  "knitr",
  "rmarkdown"
)

optional_packages <- c("plotly", "htmlwidgets")

missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]

if (length(missing_packages) > 0) {
  stop(
    paste(
      "Install these packages before running the full workflow:",
      paste(missing_packages, collapse = ", ")
    )
  )
}

invisible(lapply(required_packages, library, character.only = TRUE))

set.seed(42)

input_path <- "../li_accidents_clean.csv"
output_dir <- "local_artifacts/output"
```

```r
run_profile <- "local_safe"

if (!run_profile %in% c("local_safe", "consensus_16gb", "full_srs")) {
  stop("run_profile must be one of 'local_safe', 'consensus_16gb', or 'full_srs'.")
}

phase6_validation_mode <- dplyr::case_when(
  run_profile == "full_srs" ~ "full",
  run_profile == "consensus_16gb" ~ "consensus_sampled",
  TRUE ~ "sampled"
)
phase6_validation_sample_n <- dplyr::case_when(
  run_profile == "full_srs" ~ NA_integer_,
  run_profile == "consensus_16gb" ~ 4000L,
  TRUE ~ 5000L
)
phase6_validation_repeats <- dplyr::case_when(
  run_profile == "consensus_16gb" ~ 4L,
  TRUE ~ 1L
)

phase6_stability_mode <- dplyr::case_when(
  run_profile == "full_srs" ~ "full",
  run_profile == "consensus_16gb" ~ "consensus_sampled",
  TRUE ~ "sampled"
)
phase6_stability_sample_n <- dplyr::case_when(
  run_profile == "full_srs" ~ NA_integer_,
  run_profile == "consensus_16gb" ~ 2500L,
  TRUE ~ 3000L
)
phase6_stability_repeats <- dplyr::case_when(
  run_profile == "consensus_16gb" ~ 3L,
  TRUE ~ 1L
)
phase6_stability_bootstraps <- dplyr::case_when(
  run_profile == "full_srs" ~ 50L,
  run_profile == "consensus_16gb" ~ 10L,
  TRUE ~ 15L
)
phase9_interactive_mode <- "auto"

interactive_exports_enabled <- dplyr::case_when(
  phase9_interactive_mode == "off" ~ FALSE,
  phase9_interactive_mode == "on" ~ all(
    vapply(optional_packages, requireNamespace, logical(1), quietly = TRUE)
  ),
  phase9_interactive_mode == "auto" ~ all(
    vapply(optional_packages, requireNamespace, logical(1), quietly = TRUE)
  ),
  TRUE ~ FALSE
)
```

```r
if (!dir.exists(output_dir)) {
  dir.create(output_dir, recursive = TRUE)
}

if (!file.exists(input_path)) {
  stop("Expected input file not found: ", input_path)
}

phase_1_check <- data.frame(
  item = c(
    "Input dataset",
    "Output directory",
    "Seed",
    "Run profile",
    "Phase 6 validation mode",
    "Phase 6 sample size",
    "Phase 6 validation repeats",
    "Phase 6 stability mode",
    "Phase 6 stability sample size",
    "Phase 6 stability repeats",
    "Phase 6 stability bootstraps",
    "Phase 9 interactive mode",
    "Working directory"
  ),
  value = c(
    normalizePath(input_path, winslash = "/"),
    normalizePath(output_dir, winslash = "/"),
    "42",
    run_profile,
    phase6_validation_mode,
    ifelse(is.na(phase6_validation_sample_n), "Full data", as.character(phase6_validation_sample_n)),
    as.character(phase6_validation_repeats),
    phase6_stability_mode,
    ifelse(is.na(phase6_stability_sample_n), "Full data", as.character(phase6_stability_sample_n)),
    as.character(phase6_stability_repeats),
    as.character(phase6_stability_bootstraps),
    ifelse(
      interactive_exports_enabled,
      "Enabled",
      ifelse(phase9_interactive_mode == "off", "Off for safe local runs", "Packages missing")
    ),
    normalizePath(".", winslash = "/")
  )
)

knitr::kable(phase_1_check, caption = "Phase 1 setup check")
```

Table 1: Phase 1 setup check

| item | value |
|---|---|
| Input dataset | /Users/survaghasiya/Documents/AMS 597 Statistical computing Project/li_accidents_clean.csv |

| item | value |
|---|---|
| Output directory | /Users/survaghasiya/Documents/AMS 597 Statistical computing Project/topic3_nyc_clustering/local_artifacts/output |
| Seed | 42 |
| Run profile | local_safe |
| Phase 6 validation mode | sampled |
| Phase 6 sample size | 5000 |
| Phase 6 validation repeats | 1 |
| Phase 6 stability mode | sampled |
| Phase 6 stability sample size | 3000 |
| Phase 6 stability repeats | 1 |
| Phase 6 stability bootstraps | 15 |
| Phase 9 interactive mode | Packages missing |
| Working directory | /Users/survaghasiya/Documents/AMS 597 Statistical computing Project/topic3_nyc_clustering |

```
df_raw <- data.table::fread(input_path, showProgress = FALSE)

data_overview <- data.frame(
  rows = nrow(df_raw),
  columns = ncol(df_raw),
  source_file = basename(input_path)
)

knitr::kable(data_overview, caption = "Input file loaded successfully")
```

Table 2: Input file loaded successfully

| rows | columns | source_file |
|---|---|---|
| 33375 | 48 | li_accidents_clean.csv |

## Phase 2: Data Ingestion and Raw Profiling

This phase stays read-only. The goal is to confirm that the cleaned Long Island dataset matches the expectations in the execution plan before any feature engineering starts.

```
expected_rows <- 33375L
expected_cols <- 48L

if (nrow(df_raw) != expected_rows || ncol(df_raw) != expected_cols) {
  stop(
    paste0(
```

```r
      "Unexpected dataset dimensions. Expected ",
      expected_rows, " rows and ", expected_cols, " columns, got ",
      nrow(df_raw), " rows and ", ncol(df_raw), " columns."
    )
  )
}

phase_2_dimensions <- data.frame(
  metric = c("Rows", "Columns"),
  expected = c(expected_rows, expected_cols),
  observed = c(nrow(df_raw), ncol(df_raw))
)

knitr::kable(phase_2_dimensions, caption = "Phase 2 dimension check")
```

Table 3: Phase 2 dimension check

| metric | expected | observed |
|---|---|---|
| Rows | 33375 | 33375 |
| Columns | 48 | 48 |

```r
county_profile <- as.data.table(df_raw)[, .N, by = County][order(match(County, c("Nassau", "Suffolk", "(
severity_profile <- as.data.table(df_raw)[, .N, by = Severity][order(as.integer(Severity))]

expected_counties <- data.table(
  County = c("Nassau", "Suffolk", "Queens"),
  expected_n = c(19390L, 13838L, 147L)
)

expected_severity <- data.table(
  Severity = c(1L, 2L, 3L, 4L),
  expected_n = c(161L, 28357L, 3938L, 919L)
)

county_check <- merge(expected_counties, county_profile, by = "County", all.x = TRUE)
severity_check <- merge(
  expected_severity,
  severity_profile[, Severity := as.integer(Severity)],
  by = "Severity",
  all.x = TRUE
)

if (any(county_check$expected_n != county_check$N)) {
  stop("County counts do not match the execution plan expectations.")
}

if (any(severity_check$expected_n != severity_check$N)) {
  stop("Severity counts do not match the execution plan expectations.")
}

fwrite(county_profile, file.path(output_dir, "phase2_county_profile.csv"))
fwrite(severity_profile, file.path(output_dir, "phase2_severity_profile.csv"))
```

```r
knitr::kable(county_check, caption = "County profile check")
```

Table 4: County profile check

| County | expected_n | N |
|---|---|---|
| Nassau | 19390 | 19390 |
| Queens | 147 | 147 |
| Suffolk | 13838 | 13838 |

```r
knitr::kable(severity_check, caption = "Severity profile check")
```

Table 5: Severity profile check

| Severity | expected_n | N |
|---|---|---|
| 1 | 161 | 161 |
| 2 | 28357 | 28357 |
| 3 | 3938 | 3938 |
| 4 | 919 | 919 |

```r
missingness_profile <- data.table(
  column = names(df_raw),
  missing_n = vapply(df_raw, function(column) sum(is.na(column)), integer(1)),
  missing_pct = round(vapply(df_raw, function(column) mean(is.na(column)) * 100, numeric(1)), 2)
)[order(-missing_pct, column)]

missingness_focus <- missingness_profile[
  column %in% c("End_Lat", "End_Lng", "Precipitation(in)", "Wind_Chill(F)")
]

fwrite(missingness_profile, file.path(output_dir, "phase2_missingness_profile.csv"))

knitr::kable(head(missingness_profile, 12), caption = "Top missingness profile")
```

Table 6: Top missingness profile

| column | missing_n | missing_pct |
|---|---|---|
| End_Lat | 13085 | 39.21 |
| End_Lng | 13085 | 39.21 |
| Precipitation(in) | 6724 | 20.15 |
| Wind_Chill(F) | 5011 | 15.01 |
| Airport_Code | 0 | 0.00 |
| Amenity | 0 | 0.00 |
| Astronomical_Twilight | 0 | 0.00 |
| Bump | 0 | 0.00 |
| City | 0 | 0.00 |
| Civil_Twilight | 0 | 0.00 |
| Country | 0 | 0.00 |

| column | missing_n | missing_pct |
|---|---|---|
| County | 0 | 0.00 |

```
knitr::kable(missingness_focus, caption = "Critical missingness checks")
```

Table 7: Critical missingness checks

| column | missing_n | missing_pct |
|---|---|---|
| End_Lat | 13085 | 39.21 |
| End_Lng | 13085 | 39.21 |
| Precipitation(in) | 6724 | 20.15 |
| Wind_Chill(F) | 5011 | 15.01 |

```
binary_flag_profile <- data.table(
  flag = c("Bump", "Roundabout", "Turning_Loop", "Traffic_Calming", "No_Exit"),
  true_count = vapply(
    df_raw[, c("Bump", "Roundabout", "Turning_Loop", "Traffic_Calming", "No_Exit")],
    function(column) sum(as.logical(column), na.rm = TRUE),
    integer(1)
  )
)

binary_flag_profile[, true_pct := round(100 * true_count / nrow(df_raw), 4)]

fwrite(binary_flag_profile, file.path(output_dir, "phase2_binary_flag_profile.csv"))

knitr::kable(binary_flag_profile, caption = "Near-zero binary flag checks")
```

Table 8: Near-zero binary flag checks

| flag | true_count | true_pct |
|---|---|---|
| Bump | 0 | 0.000 |
| Roundabout | 0 | 0.000 |
| Turning_Loop | 0 | 0.000 |
| Traffic_Calming | 4 | 0.012 |
| No_Exit | 5 | 0.015 |

```
duration_values <- as.numeric(df_raw$Duration_Minutes)
distance_values <- as.numeric(df_raw[["Distance(mi)"]])

duration_profile <- data.table(
  metric = c(
    "Median duration",
    "Mean duration",
    "Max duration",
    "Rows above 1 day",
    "Rows above 1 week",
    "99th percentile"
```

```r
  ),
  value = c(
    median(duration_values, na.rm = TRUE),
    mean(duration_values, na.rm = TRUE),
    max(duration_values, na.rm = TRUE),
    sum(duration_values > 1440, na.rm = TRUE),
    sum(duration_values > 10080, na.rm = TRUE),
    as.numeric(quantile(duration_values, 0.99, na.rm = TRUE))
  )
)

distance_profile <- data.table(
  metric = c("Zero-distance rows", "Zero-distance pct", "Max distance"),
  value = c(
    sum(distance_values == 0, na.rm = TRUE),
    100 * mean(distance_values == 0, na.rm = TRUE),
    max(distance_values, na.rm = TRUE)
  )
)

fwrite(duration_profile, file.path(output_dir, "phase2_duration_profile.csv"))
fwrite(distance_profile, file.path(output_dir, "phase2_distance_profile.csv"))

knitr::kable(duration_profile, caption = "Duration anomaly profile")
```

Table 9: Duration anomaly profile

| metric | value |
|---|---|
| Median duration | 7.783333e+01 |
| Mean duration | 4.485395e+02 |
| Max duration | 2.236406e+06 |
| Rows above 1 day | 2.100000e+01 |
| Rows above 1 week | 6.000000e+00 |
| 99th percentile | 4.805410e+02 |

```r
knitr::kable(distance_profile, caption = "Distance profile")
```

Table 10: Distance profile

| metric | value |
|---|---|
| Zero-distance rows | 11724.00000 |
| Zero-distance pct | 35.12809 |
| Max distance | 35.62700 |

```r
wind_direction_profile <- as.data.table(df_raw)[, .N, by = Wind_Direction][order(-N, Wind_Direction)]
weather_profile <- as.data.table(df_raw)[, .N, by = Weather_Condition][order(-N, Weather_Condition)]

wind_focus <- wind_direction_profile[
  Wind_Direction %in% c("N", "North", "S", "South", "E", "East", "W", "West")
```

```r
]

fwrite(wind_direction_profile, file.path(output_dir, "phase2_wind_direction_profile.csv"))
fwrite(weather_profile, file.path(output_dir, "phase2_weather_profile.csv"))

knitr::kable(wind_focus, caption = "Wind direction duplicate label check")
```

Table 11: Wind direction duplicate label check

| Wind_Direction | N |
|---|---|
| S | 2787 |
| N | 1930 |
| W | 1540 |
| E | 836 |
| South | 757 |
| West | 675 |
| North | 625 |
| East | 319 |

```r
knitr::kable(head(weather_profile, 12), caption = "Top weather labels")
```

Table 12: Top weather labels

| Weather_Condition | N |
|---|---|
| Fair | 10224 |
| Mostly Cloudy | 5724 |
| Cloudy | 4310 |
| Partly Cloudy | 3217 |
| Clear | 2288 |
| Light Rain | 1930 |
| Overcast | 1718 |
| Scattered Clouds | 691 |
| Fog | 577 |
| Rain | 447 |
| Light Snow | 358 |
| Fair / Windy | 288 |

```r
profiling_summary <- data.table(
  finding = c(
    "Dimensions match the execution plan",
    "County counts match the Long Island scope",
    "Severity counts match the execution plan",
    "End_Lat and End_Lng remain heavily missing",
    "Precipitation and Wind_Chill remain too sparse for core clustering use",
    "Five binary roadway flags carry no signal",
    "Duration still shows extreme right-tail outliers",
    "Distance includes many zero values",
    "Wind direction labels need normalization",
    "Weather labels still need grouping before clustering"
```

```
  ),
  status = c(
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed"
  )
)

fwrite(profiling_summary, file.path(output_dir, "phase2_profiling_summary.csv"))

knitr::kable(profiling_summary, caption = "Phase 2 profiling summary")
```

Table 13: Phase 2 profiling summary

| finding | status |
|---|---|
| Dimensions match the execution plan | Confirmed |
| County counts match the Long Island scope | Confirmed |
| Severity counts match the execution plan | Confirmed |
| End_Lat and End_Lng remain heavily missing | Confirmed |
| Precipitation and Wind_Chill remain too sparse for core clustering use | Confirmed |
| Five binary roadway flags carry no signal | Confirmed |
| Duration still shows extreme right-tail outliers | Confirmed |
| Distance includes many zero values | Confirmed |
| Wind direction labels need normalization | Confirmed |
| Weather labels still need grouping before clustering | Confirmed |

## Phase 3: Feature Engineering

This phase derives the analysis-ready fields needed for the clustering inputs while keeping the original cleaned dataset unchanged in `df_raw`.

```
df_phase3 <- as.data.table(copy(df_raw))
```

```
df_phase3[, Start_Time_Parsed := lubridate::ymd_hms(Start_Time, quiet = TRUE)]

df_phase3[, hour := lubridate::hour(Start_Time_Parsed)]
df_phase3[, month := lubridate::month(Start_Time_Parsed)]
df_phase3[, weekday := factor(
  weekdays(as.Date(Start_Time_Parsed)),
  levels = ordered_weekdays,
  ordered = TRUE
)]
df_phase3[, weekend := weekdays(as.Date(Start_Time_Parsed)) %in% c("Saturday", "Sunday")]
```

```r
df_phase3[, hour_band := dplyr::case_when(
  hour >= 0 & hour <= 5 ~ "Early_AM",
  hour >= 6 & hour <= 9 ~ "AM_Rush",
  hour >= 10 & hour <= 14 ~ "Midday",
  hour >= 15 & hour <= 18 ~ "PM_Rush",
  hour >= 19 & hour <= 21 ~ "Evening",
  hour >= 22 & hour <= 23 ~ "Late_Night",
  TRUE ~ NA_character_
)]
df_phase3[, hour_band := factor(
  hour_band,
  levels = c("Early_AM", "AM_Rush", "Midday", "PM_Rush", "Evening", "Late_Night"),
  ordered = TRUE
)]
df_phase3[, season := dplyr::case_when(
  month %in% c(12L, 1L, 2L) ~ "Winter",
  month %in% c(3L, 4L, 5L) ~ "Spring",
  month %in% c(6L, 7L, 8L) ~ "Summer",
  month %in% c(9L, 10L, 11L) ~ "Fall",
  TRUE ~ NA_character_
)]
df_phase3[, season := factor(
  season,
  levels = c("Winter", "Spring", "Summer", "Fall"),
  ordered = TRUE
)]
df_phase3[, rush_hour := hour %in% c(6L, 7L, 8L, 9L, 15L, 16L, 17L, 18L)]

temporal_na_check <- data.table(
  feature = c("Start_Time_Parsed", "hour", "month", "weekday", "weekend", "hour_band", "season", "rush_h
  missing_n = c(
    sum(is.na(df_phase3$Start_Time_Parsed)),
    sum(is.na(df_phase3$hour)),
    sum(is.na(df_phase3$month)),
    sum(is.na(df_phase3$weekday)),
    sum(is.na(df_phase3$weekend)),
    sum(is.na(df_phase3$hour_band)),
    sum(is.na(df_phase3$season)),
    sum(is.na(df_phase3$rush_hour))
  )
)

if (any(temporal_na_check$missing_n > 0)) {
  stop("Phase 3 temporal feature derivation created unexpected missing values.")
}

fwrite(temporal_na_check, file.path(output_dir, "phase3_temporal_check.csv"))

knitr::kable(temporal_na_check, caption = "Phase 3 temporal feature check")
```

Table 14: Phase 3 temporal feature check

| feature | missing_n |
|---|---|
| Start_Time_Parsed | 0 |
| hour | 0 |
| month | 0 |
| weekday | 0 |
| weekend | 0 |
| hour_band | 0 |
| season | 0 |
| rush_hour | 0 |

```
duration_values <- as.numeric(df_phase3$Duration_Minutes)
duration_cap_99 <- as.numeric(quantile(duration_values, 0.99, na.rm = TRUE))
df_phase3[, Duration_cap := pmin(duration_values, duration_cap_99)]
df_phase3[, log_Duration := log1p(Duration_cap)]

duration_engineering_summary <- data.table(
  metric = c("99th percentile cap", "Median log_Duration", "Max log_Duration"),
  value = c(
    duration_cap_99,
    median(df_phase3$log_Duration, na.rm = TRUE),
    max(df_phase3$log_Duration, na.rm = TRUE)
  )
)

fwrite(duration_engineering_summary, file.path(output_dir, "phase3_duration_engineering.csv"))

knitr::kable(duration_engineering_summary, caption = "Phase 3 duration treatment")
```

Table 15: Phase 3 duration treatment

| metric | value |
|---|---|
| 99th percentile cap | 480.541000 |
| Median log_Duration | 4.367336 |
| Max log_Duration | 6.176991 |

```
weather_counts <- as.data.table(df_phase3)[, .N, by = Weather_Condition]
rare_weather <- weather_counts[N < 30, Weather_Condition]

df_phase3[, weather_group := dplyr::case_when(
  Weather_Condition %in% weather_group_map$Clear ~ "Clear",
  Weather_Condition %in% weather_group_map$Cloudy ~ "Cloudy",
  Weather_Condition %in% weather_group_map$Rain ~ "Rain",
  Weather_Condition %in% weather_group_map$Snow_Ice ~ "Snow_Ice",
  Weather_Condition %in% weather_group_map$Fog_Haze ~ "Fog_Haze",
  grepl("Thunder", Weather_Condition, ignore.case = TRUE) ~ "Thunderstorm",
  Weather_Condition %in% rare_weather ~ "Other",
  TRUE ~ "Other"
)]
```

```r
df_phase3[, weather_group := factor(
  weather_group,
  levels = c("Clear", "Cloudy", "Rain", "Snow_Ice", "Fog_Haze", "Thunderstorm", "Other")
)]

weather_group_profile <- as.data.table(df_phase3)[, .N, by = weather_group][order(-N)]
fwrite(weather_group_profile, file.path(output_dir, "phase3_weather_group_profile.csv"))

knitr::kable(weather_group_profile, caption = "Phase 3 weather group profile")
```

Table 16: Phase 3 weather group profile

| weather_group | N |
|---|---|
| Cloudy | 16193 |
| Clear | 12800 |
| Rain | 2849 |
| Fog_Haze | 624 |
| Snow_Ice | 483 |
| Other | 267 |
| Thunderstorm | 159 |

```r
df_phase3[, wind_dir_norm := normalize_wind_direction(Wind_Direction)]

wind_direction_summary <- as.data.table(df_phase3)[, .N, by = wind_dir_norm][order(-N)]

if (length(levels(df_phase3$wind_dir_norm)) > 9) {
  stop("Phase 3 wind direction normalization produced too many levels.")
}

fwrite(wind_direction_summary, file.path(output_dir, "phase3_wind_direction_normalized.csv"))

knitr::kable(wind_direction_summary, caption = "Phase 3 normalized wind direction profile")
```

Table 17: Phase 3 normalized wind direction profile

| wind_dir_norm | N |
|---|---|
| Variable_Calm | 16225 |
| S | 3544 |
| N | 2555 |
| NW | 2414 |
| SW | 2386 |
| W | 2215 |
| NE | 1880 |
| E | 1155 |
| SE | 1001 |

```r
zero_variance_flags <- c("Bump", "Roundabout", "Turning_Loop", "Traffic_Calming", "No_Exit")
near_zero_flags <- c("Give_Way", "Station", "Railway", "Amenity")
retained_flags <- c("Crossing", "Junction", "Traffic_Signal", "Stop")
```

```r
binary_audit <- data.table(
  flag = c(zero_variance_flags, near_zero_flags, retained_flags),
  true_count = vapply(
    df_phase3[, c(zero_variance_flags, near_zero_flags, retained_flags), with = FALSE],
    function(column) sum(as.logical(column), na.rm = TRUE),
    integer(1)
  )
)

binary_audit[, true_pct := round(100 * true_count / nrow(df_phase3), 4)]
binary_audit[, decision := c(
  rep("Exclude: zero variance", length(zero_variance_flags)),
  rep("Exclude: near-zero variance", length(near_zero_flags)),
  rep("Retain for scenario matrix", length(retained_flags))
)]

for (flag_name in retained_flags) {
  df_phase3[, (flag_name) := factor(
    as.logical(get(flag_name)),
    levels = c(FALSE, TRUE),
    labels = c("FALSE", "TRUE")
  )]
}

fwrite(binary_audit, file.path(output_dir, "phase3_binary_audit.csv"))

knitr::kable(binary_audit, caption = "Phase 3 binary flag audit")
```

Table 18: Phase 3 binary flag audit

| flag | true_count | true_pct | decision |
|---|---|---|---|
| Bump | 0 | 0.0000 | Exclude: zero variance |
| Roundabout | 0 | 0.0000 | Exclude: zero variance |
| Turning_Loop | 0 | 0.0000 | Exclude: zero variance |
| Traffic_Calming | 4 | 0.0120 | Exclude: zero variance |
| No_Exit | 5 | 0.0150 | Exclude: zero variance |
| Give_Way | 304 | 0.9109 | Exclude: near-zero variance |
| Station | 233 | 0.6981 | Exclude: near-zero variance |
| Railway | 50 | 0.1498 | Exclude: near-zero variance |
| Amenity | 112 | 0.3356 | Exclude: near-zero variance |
| Crossing | 3374 | 10.1094 | Retain for scenario matrix |
| Junction | 5332 | 15.9760 | Retain for scenario matrix |
| Traffic_Signal | 3186 | 9.5461 | Retain for scenario matrix |
| Stop | 347 | 1.0397 | Retain for scenario matrix |

```r
phase_3_summary <- data.table(
  feature_group = c(
    "Temporal features",
    "Duration treatment",
    "Weather grouping",
    "Wind normalization",
```

```
    "Binary flag audit"
  ),
  status = c(
    "Added hour, hour_band, weekday, weekend, month, season, rush_hour",
    "Added Duration_cap and log_Duration",
    "Added weather_group with 7 levels",
    "Added wind_dir_norm with at most 9 levels",
    "Marked zero and near-zero flags for exclusion; retained core road-context flags"
  )
)

fwrite(phase_3_summary, file.path(output_dir, "phase3_summary.csv"))

knitr::kable(phase_3_summary, caption = "Phase 3 feature engineering summary")
```

Table 19: Phase 3 feature engineering summary

| feature_group | status |
|---|---|
| Temporal features | Added hour, hour_band, weekday, weekend, month, season, rush_hour |
| Duration treatment | Added Duration_cap and log_Duration |
| Weather grouping | Added weather_group with 7 levels |
| Wind normalization | Added wind_dir_norm with at most 9 levels |
| Binary flag audit | Marked zero and near-zero flags for exclusion; retained core road-context flags |

## Phase 4: Build Clustering Input Matrices

This phase constructs the final matrices that will be used by the clustering methods. Nothing outside these matrices should enter the clustering algorithms.

```
df_scenario_base <- as.data.table(copy(df_phase3))

df_scenario_base[, weekend := factor(
  weekend,
  levels = c(FALSE, TRUE),
  labels = c("FALSE", "TRUE")
)]
df_scenario_base[, rush_hour := factor(
  rush_hour,
  levels = c(FALSE, TRUE),
  labels = c("FALSE", "TRUE")
)]
df_scenario_base[, Sunrise_Sunset := factor(Sunrise_Sunset)]

scenario_factor_cols <- c(
  "hour_band", "weekday", "weekend", "season", "rush_hour",
  "Sunrise_Sunset", "weather_group", "Crossing", "Junction",
  "Traffic_Signal", "Stop"
)

scenario_numeric_cols <- c(
```

```r
  "Temperature(F)", "Humidity(%)", "Pressure(in)",
  "Visibility(mi)", "Wind_Speed(mph)"
)

if (mean(is.na(df_scenario_base$log_Duration)) < 0.05) {
  scenario_numeric_cols <- c(scenario_numeric_cols, "log_Duration")
}

scenario_columns <- c(scenario_factor_cols, scenario_numeric_cols)

df_scenario <- df_scenario_base[, ..scenario_columns]

excluded_columns <- c(
  "Severity", "ID", "Source", "Description", "Street", "City", "Zipcode", "nzip",
  "Start_Lat", "Start_Lng", "End_Lat", "End_Lng", "County", "Distance(mi)",
  "Precipitation(in)", "Wind_Chill(F)", "Bump", "Roundabout", "Turning_Loop",
  "Traffic_Calming", "No_Exit", "Give_Way", "Station", "Railway", "Amenity"
)

unexpected_excluded <- intersect(names(df_scenario), excluded_columns)
if (length(unexpected_excluded) > 0) {
  stop(
    paste(
      "Excluded columns were found in df_scenario:",
      paste(unexpected_excluded, collapse = ", ")
    )
  )
}

numeric_imputation_log <- data.table(
  column = character(),
  missing_before = integer(),
  missing_pct = numeric(),
  imputation = character()
)

for (column_name in scenario_numeric_cols) {
  missing_before <- sum(is.na(df_scenario[[column_name]]))
  if (!missing_before) {
    next
  }

  missing_pct <- 100 * mean(is.na(df_scenario[[column_name]]))
  median_value <- median(df_scenario[[column_name]], na.rm = TRUE)
  df_scenario[is.na(get(column_name)), (column_name) := median_value]

  numeric_imputation_log <- rbind(
    numeric_imputation_log,
    data.table(
      column = column_name,
      missing_before = missing_before,
      missing_pct = round(missing_pct, 4),
      imputation = "Median"
```

```r
    )
  )
}

factor_imputation_log <- data.table(
  column = character(),
  missing_before = integer(),
  rule_applied = character()
)

for (column_name in scenario_factor_cols) {
  missing_before <- sum(is.na(df_scenario[[column_name]]))
  if (!missing_before) {
    next
  }

  if (missing_before < 10) {
    fill_value <- mode_value(df_scenario[[column_name]])
    df_scenario[is.na(get(column_name)), (column_name) := fill_value]
    rule_applied <- "Mode"
  } else {
    df_scenario[[column_name]] <- addNA(df_scenario[[column_name]])
    levels(df_scenario[[column_name]])[is.na(levels(df_scenario[[column_name]]))] <- "Unknown"
    rule_applied <- "Unknown level"
  }

  factor_imputation_log <- rbind(
    factor_imputation_log,
    data.table(
      column = column_name,
      missing_before = missing_before,
      rule_applied = rule_applied
    )
  )
}

scenario_na_total <- sum(is.na(df_scenario))
if (scenario_na_total > 0) {
  stop("df_scenario still contains missing values after Phase 4 imputations.")
}

scenario_column_profile <- data.table(
  column = names(df_scenario),
  type = vapply(df_scenario, function(column) class(column)[1], character(1))
)

scenario_matrix_summary <- data.table(
  metric = c("Rows", "Columns", "Factor columns", "Numeric columns", "Total NAs"),
  value = c(
    nrow(df_scenario),
    ncol(df_scenario),
    length(scenario_factor_cols),
    length(scenario_numeric_cols),
```

```r
    scenario_na_total
  )
)

fwrite(scenario_column_profile, file.path(output_dir, "phase4_scenario_columns.csv"))
fwrite(numeric_imputation_log, file.path(output_dir, "phase4_numeric_imputation_log.csv"))
fwrite(factor_imputation_log, file.path(output_dir, "phase4_factor_imputation_log.csv"))
fwrite(scenario_matrix_summary, file.path(output_dir, "phase4_scenario_matrix_summary.csv"))

knitr::kable(scenario_matrix_summary, caption = "Phase 4 scenario matrix summary")
```

Table 20: Phase 4 scenario matrix summary

| metric | value |
|---|---|
| Rows | 33375 |
| Columns | 17 |
| Factor columns | 11 |
| Numeric columns | 6 |
| Total NAs | 0 |

```r
knitr::kable(scenario_column_profile, caption = "Phase 4 scenario column list")
```

Table 21: Phase 4 scenario column list

| column | type |
|---|---|
| hour_band | ordered |
| weekday | ordered |
| weekend | factor |
| season | ordered |
| rush_hour | factor |
| Sunrise_Sunset | factor |
| weather_group | factor |
| Crossing | factor |
| Junction | factor |
| Traffic_Signal | factor |
| Stop | factor |
| Temperature(F) | numeric |
| Humidity(%) | integer |
| Pressure(in) | numeric |
| Visibility(mi) | numeric |
| Wind_Speed(mph) | numeric |
| log_Duration | numeric |

```r
if (any(is.na(df_phase3$Start_Lat)) || any(is.na(df_phase3$Start_Lng))) {
  stop("Start_Lat or Start_Lng contains missing values, so df_spatial cannot be built.")
}

lat0 <- 40.8
lng0 <- -73.2
```

```r
df_spatial <- data.table(
  x_km = (df_phase3$Start_Lng - lng0) * 111.32 * cos(lat0 * pi / 180),
  y_km = (df_phase3$Start_Lat - lat0) * 110.574
)

spatial_extent <- data.table(
  metric = c("Rows", "Columns", "x_km min", "x_km max", "x_km span", "y_km min", "y_km max", "y_km span
  value = c(
    nrow(df_spatial),
    ncol(df_spatial),
    min(df_spatial$x_km),
    max(df_spatial$x_km),
    diff(range(df_spatial$x_km)),
    min(df_spatial$y_km),
    max(df_spatial$y_km),
    diff(range(df_spatial$y_km))
  )
)

if (sum(is.na(df_spatial)) > 0) {
  stop("df_spatial contains missing values after coordinate projection.")
}

fwrite(spatial_extent, file.path(output_dir, "phase4_spatial_extent.csv"))

knitr::kable(spatial_extent, caption = "Phase 4 spatial matrix extent")
```

Table 22: Phase 4 spatial matrix extent

| metric | value |
|---|---:|
| Rows | 33375.00000 |
| Columns | 2.00000 |
| x_km min | -46.06598 |
| x_km max | 90.43842 |
| x_km span | 136.50441 |
| y_km min | -23.82284 |
| y_km max | 37.95928 |
| y_km span | 61.78212 |

```r
phase_4_summary <- data.table(
  matrix = c("df_scenario", "df_spatial"),
  rows = c(nrow(df_scenario), nrow(df_spatial)),
  columns = c(ncol(df_scenario), ncol(df_spatial)),
  na_count = c(sum(is.na(df_scenario)), sum(is.na(df_spatial)))
)

fwrite(phase_4_summary, file.path(output_dir, "phase4_summary.csv"))

knitr::kable(phase_4_summary, caption = "Phase 4 matrix readiness summary")
```

Table 23: Phase 4 matrix readiness summary

| matrix | rows | columns | na_count |
|---|---|---|---|
| df_scenario | 33375 | 17 | 0 |
| df_spatial | 33375 | 2 | 0 |

## Phase 5: k-Prototypes Scenario Clustering

This phase runs the candidate mixed-type clustering solutions on `df_scenario` and stores the diagnostics needed for later model selection.

```
scenario_type_check <- data.table(
  column = names(df_scenario),
  class = vapply(df_scenario, function(column) class(column)[1], character(1))
)

factor_check <- scenario_type_check[column %in% scenario_factor_cols]
numeric_check <- scenario_type_check[column %in% scenario_numeric_cols]

if (!all(vapply(df_scenario[, ..scenario_factor_cols], is.factor, logical(1)))) {
  stop("One or more Phase 5 scenario factor columns are not stored as factors.")
}

if (!all(vapply(df_scenario[, ..scenario_numeric_cols], is.numeric, logical(1)))) {
  stop("One or more Phase 5 scenario numeric columns are not stored as numeric values.")
}

fwrite(scenario_type_check, file.path(output_dir, "phase5_scenario_type_check.csv"))

knitr::kable(scenario_type_check, caption = "Phase 5 scenario type check")
```

Table 24: Phase 5 scenario type check

| column | class |
|---|---|
| hour_band | ordered |
| weekday | ordered |
| weekend | factor |
| season | ordered |
| rush_hour | factor |
| Sunrise_Sunset | factor |
| weather_group | factor |
| Crossing | factor |
| Junction | factor |
| Traffic_Signal | factor |
| Stop | factor |
| Temperature(F) | numeric |
| Humidity(%) | integer |
| Pressure(in) | numeric |
| Visibility(mi) | numeric |
| Wind_Speed(mph) | numeric |

| column | class |
|---|---|
| log_Duration | numeric |

```r
candidate_k <- 3:8
kp_results <- list()

for (k_value in candidate_k) {
  set.seed(42)
  kp_results[[as.character(k_value)]] <- clustMixType::kproto(
    x = df_scenario,
    k = k_value,
    lambda = NULL,
    nstart = 10,
    type = "gower",
    keep.data = TRUE,
    verbose = FALSE
  )
}
```

```r
cluster_fit_summary <- rbindlist(
  lapply(names(kp_results), function(k_label) {
    model_fit <- kp_results[[k_label]]
    cluster_sizes <- table(model_fit$cluster)

    data.table(
      k = as.integer(k_label),
      tot_withinss = model_fit$tot.withinss,
      lambda = if (is.null(model_fit$lambda)) NA_real_ else as.numeric(model_fit$lambda)[1],
      min_cluster_size = min(cluster_sizes),
      max_cluster_size = max(cluster_sizes),
      n_clusters = length(cluster_sizes),
      potentially_unstable = min(cluster_sizes) < 200
    )
  })
)

cluster_size_table <- rbindlist(
  lapply(names(kp_results), function(k_label) {
    model_fit <- kp_results[[k_label]]
    data.table(
      k = as.integer(k_label),
      cluster = names(table(model_fit$cluster)),
      size = as.integer(table(model_fit$cluster))
    )
  })
)

dominance_checks <- rbindlist(
  lapply(names(kp_results), function(k_label) {
    model_fit <- kp_results[[k_label]]
    clustered_data <- copy(df_scenario)
    clustered_data[, cluster := factor(model_fit$cluster)]
```

```r
    rbindlist(lapply(scenario_factor_cols, function(column_name) {
      clustered_data[, .(
        dominant_share = round(dominant_share(get(column_name)), 4)
      ), by = cluster][
        , `:=`(
          k = as.integer(k_label),
          feature = column_name
        )
      ]
    }), fill = TRUE)
  }),
  fill = TRUE
)

dominance_flags <- dominance_checks[
  dominant_share >= 0.9
][order(k, cluster, feature)]

fwrite(cluster_fit_summary, file.path(output_dir, "phase5_kproto_fit_summary.csv"))
fwrite(cluster_size_table, file.path(output_dir, "phase5_kproto_cluster_sizes.csv"))
fwrite(dominance_checks, file.path(output_dir, "phase5_kproto_dominance_checks.csv"))
fwrite(dominance_flags, file.path(output_dir, "phase5_kproto_dominance_flags.csv"))

knitr::kable(cluster_fit_summary, caption = "Phase 5 k-prototypes fit summary")
```

Table 25: Phase 5 k-prototypes fit summary

| k | tot_withinss | lambda | min_cluster_size | max_cluster_size | n_clusters | potentially_unstable |
|---|---|---|---|---|---|---|
| 3 | 85386.21 | NA | 9312 | 13470 | 3 | FALSE |
| 4 | 80677.86 | NA | 5126 | 10584 | 4 | FALSE |
| 5 | 78654.23 | NA | 2319 | 10509 | 5 | FALSE |
| 6 | 75186.66 | NA | 2562 | 8994 | 6 | FALSE |
| 7 | 72539.36 | NA | 2089 | 9126 | 7 | FALSE |
| 8 | 69535.25 | NA | 2137 | 8543 | 8 | FALSE |

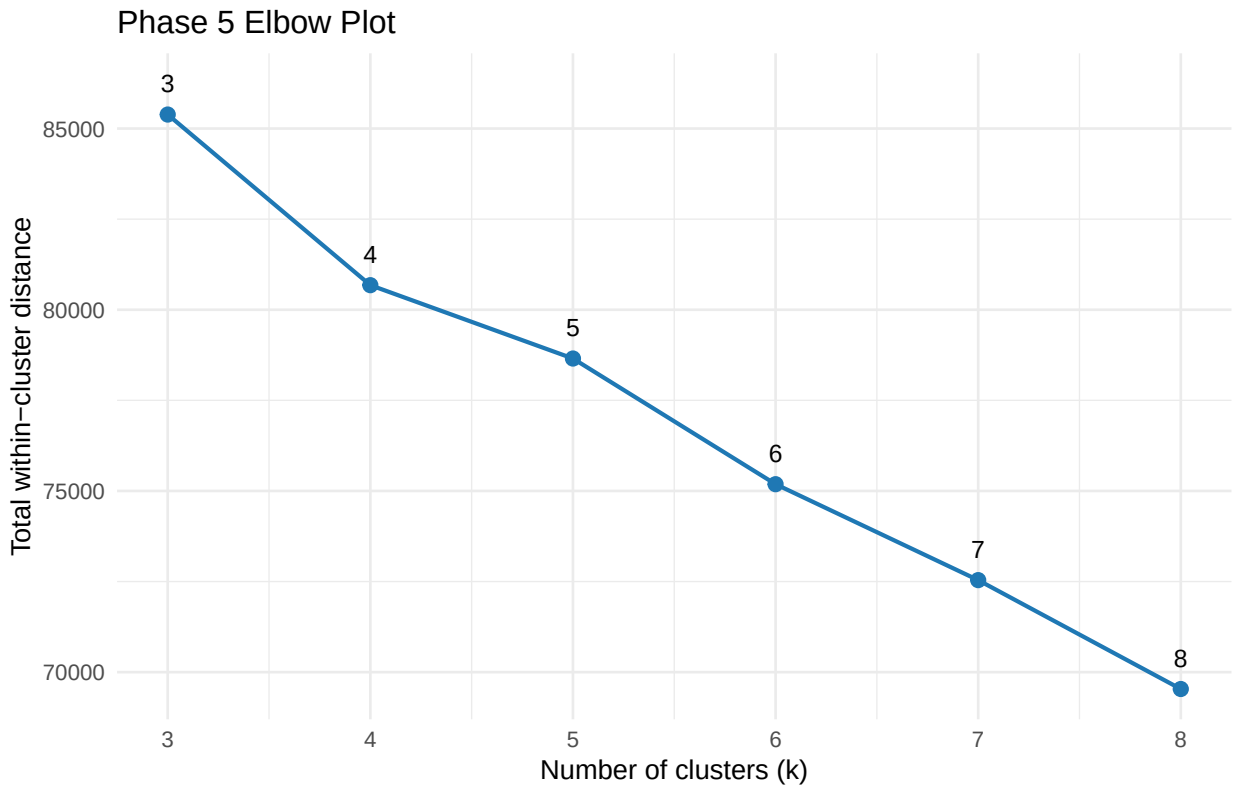
```r
knitr::kable(cluster_size_table, caption = "Phase 5 cluster sizes by k")
```

Table 26: Phase 5 cluster sizes by k

| k | cluster | size |
|---|---|---|
| 3 | 1 | 10593 |
| 3 | 2 | 13470 |
| 3 | 3 | 9312 |
| 4 | 1 | 10584 |
| 4 | 2 | 10353 |
| 4 | 3 | 7312 |
| 4 | 4 | 5126 |
| 5 | 1 | 10389 |
| 5 | 2 | 3857 |
| 5 | 3 | 10509 |

| k | cluster | size |
|---|---|---|
| 5 | 4 | 6301 |
| 5 | 5 | 2319 |
| 6 | 1 | 6439 |
| 6 | 2 | 8994 |
| 6 | 3 | 3800 |
| 6 | 4 | 2932 |
| 6 | 5 | 8648 |
| 6 | 6 | 2562 |
| 7 | 1 | 3516 |
| 7 | 2 | 9126 |
| 7 | 3 | 6395 |
| 7 | 4 | 6410 |
| 7 | 5 | 2089 |
| 7 | 6 | 2803 |
| 7 | 7 | 3036 |
| 8 | 1 | 2137 |
| 8 | 2 | 3566 |
| 8 | 3 | 4281 |
| 8 | 4 | 8543 |
| 8 | 5 | 2293 |
| 8 | 6 | 4051 |
| 8 | 7 | 2915 |
| 8 | 8 | 5589 |

```r
elbow_plot <- ggplot2::ggplot(
  cluster_fit_summary,
  ggplot2::aes(x = k, y = tot_withinss)
) +
  ggplot2::geom_line(color = "#1f78b4", linewidth = 0.8) +
  ggplot2::geom_point(color = "#1f78b4", size = 2.4) +
  ggplot2::geom_text(
    ggplot2::aes(label = k),
    nudge_y = max(cluster_fit_summary$tot_withinss) * 0.01,
    size = 3.5
  ) +
  ggplot2::labs(
    title = "Phase 5 Elbow Plot",
    x = "Number of clusters (k)",
    y = "Total within-cluster distance"
  ) +
  ggplot2::theme_minimal()

print(elbow_plot)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase5_kproto_elbow_plot.png"),
  plot = elbow_plot,
  width = 7,
  height = 4.5,
  dpi = 300
)
```

```r
phase_5_summary <- data.table(
  checkpoint = c(
    "Candidate solutions created",
    "k range covered",
    "Elbow plot saved",
    "Cluster sizes recorded",
    "Dominance checks recorded"
  ),
  status = c(
    length(kp_results) == 6,
    paste(candidate_k, collapse = ", "),
    file.exists(file.path(output_dir, "phase5_kproto_elbow_plot.png")),
    nrow(cluster_size_table) > 0,
    nrow(dominance_checks) > 0
  )
)

fwrite(phase_5_summary, file.path(output_dir, "phase5_summary.csv"))

knitr::kable(phase_5_summary, caption = "Phase 5 readiness summary")
```

Table 27: Phase 5 readiness summary

| checkpoint | status |
|---|---|
| Candidate solutions created | TRUE |
| k range covered | 3, 4, 5, 6, 7, 8 |
| Elbow plot saved | TRUE |
| Cluster sizes recorded | TRUE |
| Dominance checks recorded | TRUE |

## Phase 6: Cluster Validation and Stability

This phase uses internal validation, bootstrap stability, and cluster profile checks to choose a defensible value of `k`.

```
validation_input <- copy(df_scenario)
validation_rows <- nrow(df_scenario)
validation_repeat_results <- NULL

if (identical(phase6_validation_mode, "consensus_sampled")) {
  validation_rows <- min(phase6_validation_sample_n, nrow(df_scenario))
  validation_repeat_results <- rbindlist(
    lapply(seq_len(phase6_validation_repeats), function(repeat_id) {
      set.seed(42 + repeat_id)
      sampled_index <- sort(sample.int(nrow(df_scenario), validation_rows))
      validation_input_repeat <- copy(df_scenario[sampled_index])

      val_repeat <- clustMixType::validation_kproto(
        method = "silhouette",
        data = validation_input_repeat,
        k = candidate_k,
        nstart = 5,
        type = "gower",
        kp_obj = "all",
        verbose = FALSE
      )

      data.table(
        repeat_id = repeat_id,
        validation_rows = validation_rows,
        repeat_k_opt = as.integer(val_repeat$k_opt),
        k = as.integer(names(val_repeat$indices)),
        silhouette = as.numeric(val_repeat$indices)
      )
    }),
    fill = TRUE
  )

  validation_optimal_counts <- unique(
    validation_repeat_results[, .(repeat_id, repeat_k_opt)]
```

```r
  )[, .(optimal_repeat_count = .N), by = .(k = repeat_k_opt)]

  validation_table <- validation_repeat_results[, .(
    silhouette = mean(silhouette),
    sd_silhouette = stats::sd(silhouette),
    median_silhouette = stats::median(silhouette)
  ), by = k]

  validation_table <- merge(
    validation_table,
    validation_optimal_counts,
    by = "k",
    all.x = TRUE
  )
  validation_table[is.na(optimal_repeat_count), optimal_repeat_count := 0L]
  validation_table[, optimal_repeat_share := round(optimal_repeat_count / phase6_validation_repeats, 4)]
  validation_table <- validation_table[order(-optimal_repeat_count, -silhouette, k)]
  validation_optimal_k <- validation_table$k[1]
} else {
  if (identical(phase6_validation_mode, "sampled")) {
    set.seed(42)
    validation_rows <- min(phase6_validation_sample_n, nrow(df_scenario))
    sampled_index <- sort(sample.int(nrow(df_scenario), validation_rows))
    validation_input <- copy(df_scenario[sampled_index])
  }

  val_result <- clustMixType::validation_kproto(
    method = "silhouette",
    data = validation_input,
    k = candidate_k,
    nstart = 5,
    type = "gower",
    kp_obj = "all",
    verbose = FALSE
  )

  validation_table <- data.table(
    k = as.integer(names(val_result$indices)),
    silhouette = as.numeric(val_result$indices),
    sd_silhouette = NA_real_,
    median_silhouette = as.numeric(val_result$indices),
    optimal_repeat_count = 1L,
    optimal_repeat_share = 1
  )[order(-silhouette, k)]
  validation_optimal_k <- as.integer(val_result$k_opt)
}

validation_metadata <- data.table(
  setting = c("validation_mode", "validation_rows", "full_rows", "sample_fraction", "validation_repeats",
  value = c(
    phase6_validation_mode,
    validation_rows,
    nrow(df_scenario),
```

```r
    round(validation_rows / nrow(df_scenario), 4),
    phase6_validation_repeats
  )
)

top_two_k <- head(validation_table$k, 2)

fwrite(validation_table, file.path(output_dir, "phase6_validation_indices.csv"))
fwrite(validation_metadata, file.path(output_dir, "phase6_validation_metadata.csv"))
if (!is.null(validation_repeat_results)) {
  fwrite(validation_repeat_results, file.path(output_dir, "phase6_validation_repeats.csv"))
}

knitr::kable(validation_table, caption = "Phase 6 internal validation indices")
```

Table 28: Phase 6 internal validation indices

| k | silhouette | sd_silhouette | median_silhouette | optimal_repeat_count | optimal_repeat_share |
|---|---|---|---|---|---|
| 5 | 0.1769223 | NA | 0.1769223 | 1 | 1 |
| 7 | 0.1615142 | NA | 0.1615142 | 1 | 1 |
| 3 | 0.1578978 | NA | 0.1578978 | 1 | 1 |
| 8 | 0.1532625 | NA | 0.1532625 | 1 | 1 |
| 6 | 0.1514728 | NA | 0.1514728 | 1 | 1 |
| 4 | 0.1492324 | NA | 0.1492324 | 1 | 1 |

```r
knitr::kable(validation_metadata, caption = "Phase 6 validation run settings")
```

Table 29: Phase 6 validation run settings

| setting | value |
|---|---|
| validation_mode | sampled |
| validation_rows | 5000 |
| full_rows | 33375 |
| sample_fraction | 0.1498 |
| validation_repeats | 1 |

```r
if (!is.null(validation_repeat_results)) {
  knitr::kable(validation_repeat_results, caption = "Phase 6 validation repeat detail")
}
```

```r
stability_input <- copy(df_scenario)
stability_rows <- nrow(df_scenario)
stability_models <- kp_results[as.character(top_two_k)]
stability_repeat_results <- NULL
stability_bootstrap <- data.table()

if (identical(phase6_stability_mode, "consensus_sampled")) {
  stability_rows <- min(phase6_stability_sample_n, nrow(df_scenario))
```

```r
consensus_stability_outputs <- lapply(seq_len(phase6_stability_repeats), function(repeat_id) {
    set.seed(420 + repeat_id)
    stability_index <- sort(sample.int(nrow(df_scenario), stability_rows))
    stability_input_repeat <- copy(df_scenario[stability_index])

    repeat_models <- lapply(top_two_k, function(k_value) {
      set.seed(420 + repeat_id)
      clustMixType::kproto(
        x = stability_input_repeat,
        k = k_value,
        lambda = NULL,
        nstart = 5,
        type = "gower",
        keep.data = TRUE,
        verbose = FALSE
      )
    })
    names(repeat_models) <- as.character(top_two_k)

    repeat_stability_outputs <- lapply(top_two_k, function(k_value) {
      stab <- clustMixType::stability_kproto(
        object = repeat_models[[as.character(k_value)]],
        method = c("rand", "jaccard"),
        B = phase6_stability_bootstraps,
        verbose = FALSE,
        nstart = 5
      )

      bootstrap_dt <- as.data.table(stab[[2]])
      bootstrap_dt[, `:=`(k = k_value, repeat_id = repeat_id)]

      list(
        summary_dt = data.table(
          repeat_id = repeat_id,
          k = k_value,
          rand = as.numeric(stab[[1]]["R"]),
          jaccard = as.numeric(stab[[1]]["J"])
        ),
        bootstrap_dt = bootstrap_dt
      )
    })

    list(
      summary_dt = rbindlist(lapply(repeat_stability_outputs, `[[`, "summary_dt"), fill = TRUE),
      bootstrap_dt = rbindlist(lapply(repeat_stability_outputs, `[[`, "bootstrap_dt"), fill = TRUE)
    )
  })

stability_repeat_results <- rbindlist(
  lapply(consensus_stability_outputs, `[[`, "summary_dt"),
  fill = TRUE
)
stability_bootstrap <- rbindlist(
```

```r
    lapply(consensus_stability_outputs, `[[`, "bootstrap_dt"),
    fill = TRUE
  )

  stability_summary <- stability_repeat_results[, .(
    rand = mean(rand),
    jaccard = mean(jaccard),
    sd_rand = stats::sd(rand),
    sd_jaccard = stats::sd(jaccard),
    stability_repeats = .N,
    stable_repeat_share = round(mean(jaccard >= 0.6), 4),
    strong_repeat_share = round(mean(jaccard >= 0.75), 4),
    stable_jaccard = mean(jaccard >= 0.6) >= 0.5,
    strong_jaccard = mean(jaccard >= 0.75) >= 0.5
  ), by = k][order(-jaccard, -rand, k)]
} else {
  if (identical(phase6_stability_mode, "sampled")) {
    set.seed(42)
    stability_rows <- min(phase6_stability_sample_n, nrow(df_scenario))
    stability_index <- sort(sample.int(nrow(df_scenario), stability_rows))
    stability_input <- copy(df_scenario[stability_index])

    stability_models <- lapply(top_two_k, function(k_value) {
      set.seed(42)
      clustMixType::kproto(
        x = stability_input,
        k = k_value,
        lambda = NULL,
        nstart = 5,
        type = "gower",
        keep.data = TRUE,
        verbose = FALSE
      )
    })
    names(stability_models) <- as.character(top_two_k)
  }

  stability_results <- lapply(top_two_k, function(k_value) {
    set.seed(42)
    stab <- clustMixType::stability_kproto(
      object = stability_models[[as.character(k_value)]],
      method = c("rand", "jaccard"),
      B = phase6_stability_bootstraps,
      verbose = FALSE,
      nstart = 5
    )

    list(
      k = k_value,
      summary = stab[[1]],
      bootstrap = as.data.table(stab[[2]])
    )
  })
```

```r
  stability_summary <- rbindlist(
    lapply(stability_results, function(result) {
      data.table(
        k = result$k,
        rand = as.numeric(result$summary["R"]),
        jaccard = as.numeric(result$summary["J"]),
        sd_rand = NA_real_,
        sd_jaccard = NA_real_,
        stability_repeats = 1L,
        stable_repeat_share = as.numeric(result$summary["J"]) >= 0.6,
        strong_repeat_share = as.numeric(result$summary["J"]) >= 0.75,
        stable_jaccard = as.numeric(result$summary["J"]) >= 0.6,
        strong_jaccard = as.numeric(result$summary["J"]) >= 0.75
      )
    })
  )[order(-jaccard, -rand, k)]

  stability_bootstrap <- rbindlist(
    lapply(stability_results, function(result) {
      bootstrap_dt <- copy(result$bootstrap)
      bootstrap_dt[, `:=`(k = result$k, repeat_id = 1L)]
      bootstrap_dt
    }),
    fill = TRUE
  )
}

stability_metadata <- data.table(
  setting = c("stability_mode", "stability_rows", "full_rows", "sample_fraction", "bootstrap_reps", "sta
  value = c(
    phase6_stability_mode,
    stability_rows,
    nrow(df_scenario),
    round(stability_rows / nrow(df_scenario), 4),
    phase6_stability_bootstraps,
    phase6_stability_repeats
  )
)

fwrite(stability_summary, file.path(output_dir, "phase6_stability_summary.csv"))
fwrite(stability_bootstrap, file.path(output_dir, "phase6_stability_bootstrap.csv"))
fwrite(stability_metadata, file.path(output_dir, "phase6_stability_metadata.csv"))
if (!is.null(stability_repeat_results)) {
  fwrite(stability_repeat_results, file.path(output_dir, "phase6_stability_repeats.csv"))
}

knitr::kable(stability_summary, caption = "Phase 6 stability summary")
```

Table 30: Phase 6 stability summary

| k | rand | jaccard | sd_rand | sd_jaccard | stability_repeats | stable_repeat_strong | strong_repeat_stable | stable_jaccard | strong_jaccard |
|---|---|---|---|---|---|---|---|---|---|
| 5 | 0.7104380 | 0.1989349 | NA | NA | 1 | FALSE | FALSE | FALSE | FALSE |
| 7 | 0.7793816 | 0.1666265 | NA | NA | 1 | FALSE | FALSE | FALSE | FALSE |

```
knitr::kable(stability_metadata, caption = "Phase 6 stability run settings")
```

Table 31: Phase 6 stability run settings

| setting | value |
|---|---|
| stability_mode | sampled |
| stability_rows | 3000 |
| full_rows | 33375 |
| sample_fraction | 0.0899 |
| bootstrap_reps | 15 |
| stability_repeats | 1 |

```
if (!is.null(stability_repeat_results)) {
  knitr::kable(stability_repeat_results, caption = "Phase 6 stability repeat detail")
}
```

```
selected_k <- validation_table$k[1]
selected_stability <- stability_summary[k == selected_k, jaccard]

if (length(selected_stability) == 1 && selected_stability < 0.5 && nrow(stability_summary) > 1) {
  selected_k <- stability_summary[order(-jaccard, -rand, k)][1, k]
}

selected_model <- kp_results[[as.character(selected_k)]]
selected_jaccard <- if (selected_k %in% stability_summary$k) {
  stability_summary[k == selected_k, round(jaccard, 4)]
} else {
  NA_real_
}

selection_summary <- data.table(
  metric = c(
    "Validation method",
    "Validation mode",
    "Validation rows",
    "Stability mode",
    "Stability rows",
    "Stability bootstraps",
    "Validation-optimal k",
    "Top two k values for stability",
    "Selected k",
    "Selected k Jaccard"
  ),
  value = c(
```

```r
    "silhouette",
    phase6_validation_mode,
    validation_rows,
    phase6_stability_mode,
    stability_rows,
    phase6_stability_bootstraps,
    validation_optimal_k,
    paste(top_two_k, collapse = ", "),
    selected_k,
    selected_jaccard
  )
)

fwrite(selection_summary, file.path(output_dir, "phase6_selection_summary.csv"))

knitr::kable(selection_summary, caption = "Phase 6 k-selection summary")
```

Table 32: Phase 6 k-selection summary

| metric | value |
|---|---|
| Validation method | silhouette |
| Validation mode | sampled |
| Validation rows | 5000 |
| Stability mode | sampled |
| Stability rows | 3000 |
| Stability bootstraps | 15 |
| Validation-optimal k | 5 |
| Top two k values for stability | 5, 7 |
| Selected k | 5 |
| Selected k Jaccard | 0.1989 |

```r
profile_plot_path <- file.path(output_dir, "phase6_cluster_profiles.pdf")
grDevices::pdf(profile_plot_path, width = 12, height = 8)
clustMixType::clprofiles(selected_model, df_scenario)
grDevices::dev.off()
```

```
## pdf
##   2
```

```r
selected_cluster_data <- copy(df_scenario)
selected_cluster_data[, cluster := factor(selected_model$cluster)]

cluster_profile_summary <- selected_cluster_data[, .(
  dominant_hour_band = mode_value(hour_band),
  dominant_weather_group = mode_value(weather_group),
  dominant_weekend = mode_value(weekend),
  dominant_sunrise_sunset = mode_value(Sunrise_Sunset),
  dominant_junction = mode_value(Junction),
  dominant_traffic_signal = mode_value(Traffic_Signal),
  mean_temperature = round(mean(`Temperature(F)`), 2),
  mean_visibility = round(mean(`Visibility(mi)`), 2),
```

```
  mean_log_duration = round(mean(log_Duration), 3)
), by = cluster]

cluster_profile_summary[, working_name := dplyr::case_when(
  dominant_weather_group == "Snow_Ice" ~ "Winter Adverse",
  dominant_sunrise_sunset == "Night" & dominant_weather_group %in% c("Fog_Haze", "Rain", "Thunderstorm")
  dominant_weekend == "TRUE" & dominant_traffic_signal == "TRUE" ~ "Weekend Surface Road",
  dominant_hour_band %in% c("AM_Rush", "PM_Rush") & dominant_junction == "TRUE" ~ "Rush Corridor",
  TRUE ~ paste("Scenario Cluster", cluster)
)]

fwrite(cluster_profile_summary, file.path(output_dir, "phase6_cluster_profile_summary.csv"))

knitr::kable(cluster_profile_summary, caption = "Phase 6 cluster profile summary")
```

Table 33: Phase 6 cluster profile summary

| cluster | dominant_hour_band | dominant_weather_group | dominant_weekend | dominant_sunrise_sunset | dominant_junction | dominant_traffic_signal | temperature | visibility | log_duration | working_name |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | AM_Rush | Cloudy | FALSE | Day | FALSE | FALSE | 60.58 | 8.74 | 4.246 | Scenario Cluster 1 |
| 3 | AM_Rush | Clear | FALSE | Day | FALSE | FALSE | 60.49 | 9.67 | 4.436 | Scenario Cluster 3 |
| 2 | Midday | Cloudy | TRUE | Day | FALSE | FALSE | 59.14 | 9.09 | 4.439 | Scenario Cluster 2 |
| 4 | Early_AM | Cloudy | FALSE | Night | FALSE | FALSE | 50.03 | 8.44 | 4.431 | Scenario Cluster 4 |
| 5 | Early_AM | Clear | TRUE | Night | FALSE | FALSE | 57.91 | 9.38 | 4.413 | Scenario Cluster 5 |

```
phase_6_summary <- data.table(
  checkpoint = c(
    "Validation indices saved",
    "Validation metadata saved",
    "Stability summary saved",
    "Stability metadata saved",
    "Selected k assigned",
    "Cluster profile PDF saved",
    "Working names assigned"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase6_validation_indices.csv")),
    file.exists(file.path(output_dir, "phase6_validation_metadata.csv")),
    file.exists(file.path(output_dir, "phase6_stability_summary.csv")),
    file.exists(file.path(output_dir, "phase6_stability_metadata.csv")),
    !is.null(selected_model),
    file.exists(file.path(output_dir, "phase6_cluster_profiles.pdf")),
    all(nzchar(cluster_profile_summary$working_name))
```

```
  )
)

fwrite(phase_6_summary, file.path(output_dir, "phase6_summary.csv"))

knitr::kable(phase_6_summary, caption = "Phase 6 readiness summary")
```

Table 34: Phase 6 readiness summary

| checkpoint | status |
|---|---|
| Validation indices saved | TRUE |
| Validation metadata saved | TRUE |
| Stability summary saved | TRUE |
| Stability metadata saved | TRUE |
| Selected k assigned | TRUE |
| Cluster profile PDF saved | TRUE |
| Working names assigned | TRUE |

## Phase 7: DBSCAN Spatial Hotspot Clustering

This phase clusters the projected Long Island coordinates with DBSCAN so the workflow can identify corridor-style hotspot zones.

```
knn_k <- 5L
dbscan_minpts_grid <- if (run_profile == "full_srs") c(15L, 25L, 40L, 50L, 75L, 100L) else c(40L, 50L, 
dbscan_eps_grid <- seq(0.3, 2.0, by = 0.1)

knn_dist <- dbscan::kNNdist(df_spatial, k = knn_k)
knn_sorted <- sort(knn_dist)
phase7_total_rows <- nrow(df_spatial)
phase7_target_cluster_n <- 8L
phase7_target_noise_pct <- 17.5
phase7_max_cluster_share_cap <- 60

knn_profile <- data.table(
  quantile = c("50%", "75%", "90%", "95%", "97.5%", "99%"),
  value_km = round(as.numeric(quantile(knn_dist, probs = c(0.5, 0.75, 0.9, 0.95, 0.975, 0.99))), 4)
)

eps_trials <- rbindlist(
  lapply(dbscan_minpts_grid, function(minPts_value) {
    rbindlist(lapply(dbscan_eps_grid, function(eps_value) {
      db_trial <- dbscan::dbscan(df_spatial, eps = eps_value, minPts = minPts_value)
      cluster_counts <- table(db_trial$cluster)
      non_noise_sizes <- cluster_counts[names(cluster_counts) != "0"]
      non_noise_clusters <- sum(as.integer(names(cluster_counts)) > 0)
      noise_pct <- round(100 * mean(db_trial$cluster == 0), 2)

      data.table(
        eps = round(eps_value, 2),
```

```
        minPts = minPts_value,
        non_noise_clusters = non_noise_clusters,
        noise_pct = noise_pct,
        max_cluster_size = if (length(non_noise_sizes)) max(as.integer(non_noise_sizes)) else 0L,
        max_cluster_share_pct = if (length(non_noise_sizes)) round(100 * max(as.integer(non_noise_sizes
        size_balance_ok = if (length(non_noise_sizes)) round(100 * max(as.integer(non_noise_sizes)) / pl
        acceptable = non_noise_clusters >= 3 & non_noise_clusters <= 15 & noise_pct >= 5 & noise_pct <=
          if (length(non_noise_sizes)) round(100 * max(as.integer(non_noise_sizes)) / phase7_total_rows
      )
    }))
  }),
  fill = TRUE
)

acceptable_trials <- eps_trials[acceptable == TRUE]

if (nrow(acceptable_trials) > 0) {
  selected_trial <- acceptable_trials[
    order(abs(non_noise_clusters - phase7_target_cluster_n), abs(noise_pct - phase7_target_noise_pct), n
  ][1]
  selection_reason <- "Accepted trial balancing cluster count, noise level, and hotspot size concentrati
} else {
  selected_trial <- eps_trials[
    order(abs(non_noise_clusters - phase7_target_cluster_n), abs(noise_pct - phase7_target_noise_pct), n
  ][1]
  selection_reason <- "Fallback trial closest to balanced cluster count, noise level, and hotspot size c
}

eps_chosen <- selected_trial$eps[[1]]
minPts_chosen <- selected_trial$minPts[[1]]

parameter_summary <- data.table(
  metric = c(
    "kNN distance k",
    "DBSCAN minPts grid",
    "DBSCAN eps grid",
    "Largest cluster share cap (%)",
    "DBSCAN minPts",
    "Chosen eps (km)",
    "Chosen non-noise clusters",
    "Chosen noise pct",
    "Chosen largest cluster share pct",
    "Selection reason"
  ),
  value = c(
    knn_k,
    paste(dbscan_minpts_grid, collapse = ", "),
    paste(range(dbscan_eps_grid), collapse = " to "),
    phase7_max_cluster_share_cap,
    minPts_chosen,
    eps_chosen,
    selected_trial$non_noise_clusters[[1]],
    selected_trial$noise_pct[[1]],
```

```
    selected_trial$max_cluster_share_pct[[1]],
    selection_reason
  )
)

fwrite(knn_profile, file.path(output_dir, "phase7_knn_profile.csv"))
fwrite(eps_trials, file.path(output_dir, "phase7_eps_trials.csv"))
fwrite(parameter_summary, file.path(output_dir, "phase7_parameter_summary.csv"))

knitr::kable(knn_profile, caption = "Phase 7 kNN distance profile")
```

Table 35: Phase 7 kNN distance profile

| quantile | value_km |
|---|---|
| 50% | 0.0001 |
| 75% | 0.0288 |
| 90% | 0.1486 |
| 95% | 0.4177 |
| 97.5% | 0.7585 |
| 99% | 1.2474 |

```
knitr::kable(eps_trials, caption = "Phase 7 DBSCAN eps trials")
```

Table 36: Phase 7 DBSCAN eps trials

| eps | minPts | non_noise_clusters | noise_pct | max_cluster_size | max_cluster_share_pct | size_balance_ok | acceptable |
|---|---|---|---|---|---|---|---|
| 0.3 | 40 | 79 | 20.17 | 6071 | 18.19 | TRUE | FALSE |
| 0.4 | 40 | 58 | 16.82 | 8353 | 25.03 | TRUE | FALSE |
| 0.5 | 40 | 50 | 14.41 | 12841 | 38.47 | TRUE | FALSE |
| 0.6 | 40 | 47 | 11.89 | 14149 | 42.39 | TRUE | FALSE |
| 0.7 | 40 | 45 | 10.33 | 14376 | 43.07 | TRUE | FALSE |
| 0.8 | 40 | 36 | 9.22 | 15020 | 45.00 | TRUE | FALSE |
| 0.9 | 40 | 29 | 7.68 | 26956 | 80.77 | FALSE | FALSE |
| 1.0 | 40 | 22 | 6.85 | 29033 | 86.99 | FALSE | FALSE |
| 1.1 | 40 | 17 | 6.10 | 29732 | 89.08 | FALSE | FALSE |
| 1.2 | 40 | 16 | 5.39 | 30130 | 90.28 | FALSE | FALSE |
| 1.3 | 40 | 15 | 4.57 | 30512 | 91.42 | FALSE | FALSE |
| 1.4 | 40 | 13 | 3.88 | 30916 | 92.63 | FALSE | FALSE |
| 1.5 | 40 | 10 | 3.57 | 31294 | 93.76 | FALSE | FALSE |
| 1.6 | 40 | 10 | 3.10 | 31687 | 94.94 | FALSE | FALSE |
| 1.7 | 40 | 10 | 2.76 | 31768 | 95.19 | FALSE | FALSE |
| 1.8 | 40 | 13 | 2.25 | 31843 | 95.41 | FALSE | FALSE |
| 1.9 | 40 | 13 | 2.00 | 31901 | 95.58 | FALSE | FALSE |
| 2.0 | 40 | 10 | 1.60 | 32042 | 96.01 | FALSE | FALSE |
| 0.3 | 50 | 75 | 23.35 | 5545 | 16.61 | TRUE | FALSE |
| 0.4 | 50 | 59 | 18.98 | 8337 | 24.98 | TRUE | FALSE |
| 0.5 | 50 | 40 | 16.82 | 9664 | 28.96 | TRUE | FALSE |
| 0.6 | 50 | 35 | 14.86 | 13754 | 41.21 | TRUE | FALSE |
| 0.7 | 50 | 30 | 13.39 | 14316 | 42.89 | TRUE | FALSE |
| 0.8 | 50 | 30 | 11.07 | 14970 | 44.85 | TRUE | FALSE |

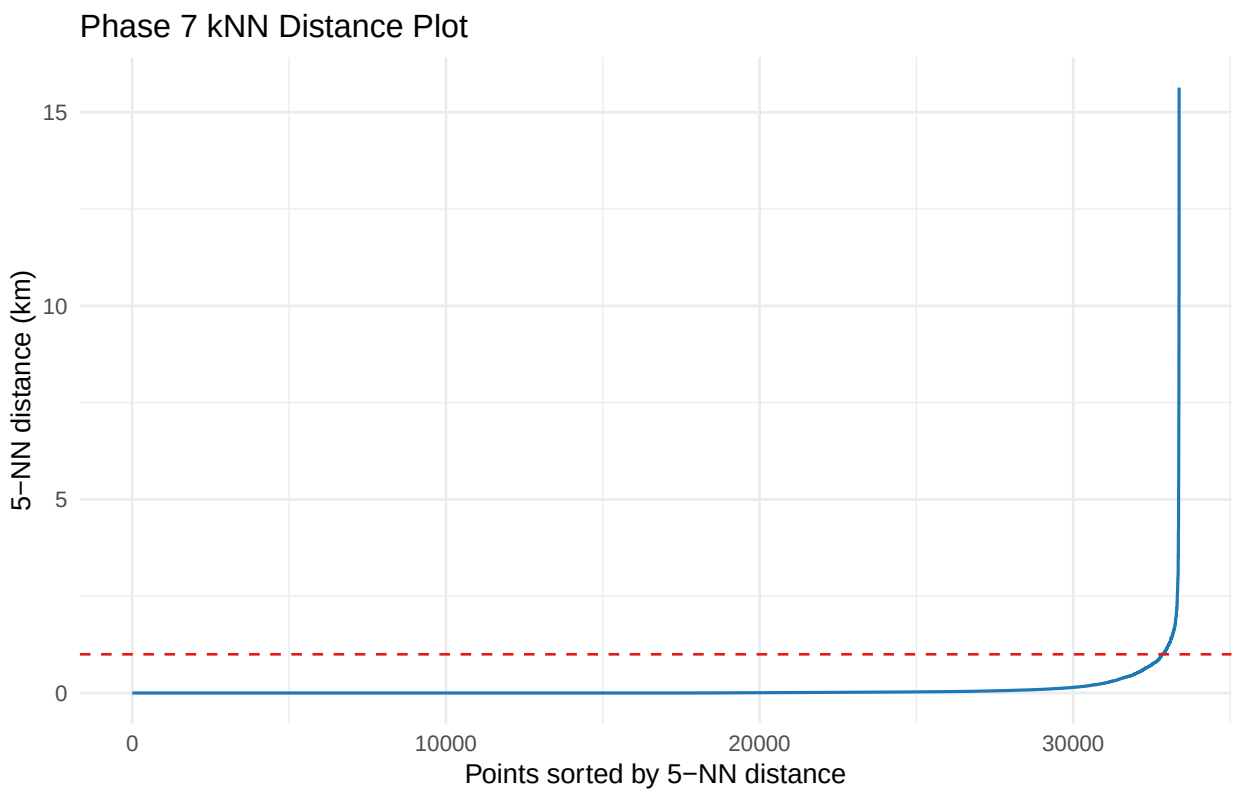
| eps | minPts | non_noise_clusters | noise_pct | max_cluster_size | max_cluster_share_pct | size_balance_ok | acceptable |
|---|---|---|---|---|---|---|---|
| 0.9 | 50 | 23 | 9.77 | 15257 | 45.71 | TRUE | FALSE |
| 1.0 | 50 | 20 | 8.25 | 27280 | 81.74 | FALSE | FALSE |
| 1.1 | 50 | 20 | 6.97 | 29138 | 87.30 | FALSE | FALSE |
| 1.2 | 50 | 16 | 6.19 | 29715 | 89.03 | FALSE | FALSE |
| 1.3 | 50 | 13 | 5.48 | 30203 | 90.50 | FALSE | FALSE |
| 1.4 | 50 | 11 | 5.02 | 30530 | 91.48 | FALSE | FALSE |
| 1.5 | 50 | 8 | 4.32 | 31027 | 92.96 | FALSE | FALSE |
| 1.6 | 50 | 9 | 3.87 | 31126 | 93.26 | FALSE | FALSE |
| 1.7 | 50 | 8 | 3.36 | 31556 | 94.55 | FALSE | FALSE |
| 1.8 | 50 | 8 | 3.01 | 31787 | 95.24 | FALSE | FALSE |
| 1.9 | 50 | 8 | 2.67 | 31861 | 95.46 | FALSE | FALSE |
| 2.0 | 50 | 8 | 2.40 | 31915 | 95.63 | FALSE | FALSE |
| 0.3 | 75 | 68 | 29.56 | 3145 | 9.42 | TRUE | FALSE |
| 0.4 | 75 | 45 | 24.81 | 7339 | 21.99 | TRUE | FALSE |
| 0.5 | 75 | 39 | 20.89 | 7943 | 23.80 | TRUE | FALSE |
| 0.6 | 75 | 25 | 19.36 | 8652 | 25.92 | TRUE | FALSE |
| 0.7 | 75 | 23 | 16.75 | 13864 | 41.54 | TRUE | FALSE |
| 0.8 | 75 | 20 | 15.32 | 14496 | 43.43 | TRUE | FALSE |
| 0.9 | 75 | 15 | 14.25 | 14750 | 44.19 | TRUE | TRUE |
| 1.0 | 75 | 13 | 13.02 | 15074 | 45.17 | TRUE | TRUE |
| 1.1 | 75 | 13 | 11.45 | 26762 | 80.19 | FALSE | FALSE |
| 1.2 | 75 | 11 | 9.42 | 27644 | 82.83 | FALSE | FALSE |
| 1.3 | 75 | 11 | 8.20 | 29582 | 88.64 | FALSE | FALSE |
| 1.4 | 75 | 10 | 7.00 | 30020 | 89.95 | FALSE | FALSE |
| 1.5 | 75 | 8 | 6.44 | 30308 | 90.81 | FALSE | FALSE |
| 1.6 | 75 | 6 | 5.88 | 30753 | 92.14 | FALSE | FALSE |
| 1.7 | 75 | 4 | 5.28 | 31053 | 93.04 | FALSE | FALSE |
| 1.8 | 75 | 4 | 4.82 | 31185 | 93.44 | FALSE | FALSE |
| 1.9 | 75 | 4 | 4.50 | 31311 | 93.82 | FALSE | FALSE |
| 2.0 | 75 | 4 | 4.04 | 31662 | 94.87 | FALSE | FALSE |
| 0.3 | 100 | 63 | 34.66 | 2289 | 6.86 | TRUE | FALSE |
| 0.4 | 100 | 49 | 28.07 | 3642 | 10.91 | TRUE | FALSE |
| 0.5 | 100 | 38 | 24.36 | 7383 | 22.12 | TRUE | FALSE |
| 0.6 | 100 | 30 | 22.04 | 8419 | 25.23 | TRUE | FALSE |
| 0.7 | 100 | 24 | 19.73 | 9584 | 28.72 | TRUE | FALSE |
| 0.8 | 100 | 19 | 17.72 | 10522 | 31.53 | TRUE | FALSE |
| 0.9 | 100 | 14 | 16.58 | 14317 | 42.90 | TRUE | TRUE |
| 1.0 | 100 | 12 | 15.22 | 14792 | 44.32 | TRUE | TRUE |
| 1.1 | 100 | 8 | 13.65 | 26644 | 79.83 | FALSE | FALSE |
| 1.2 | 100 | 6 | 12.52 | 27131 | 81.29 | FALSE | FALSE |
| 1.3 | 100 | 5 | 11.77 | 27529 | 82.48 | FALSE | FALSE |
| 1.4 | 100 | 5 | 10.26 | 27940 | 83.72 | FALSE | FALSE |
| 1.5 | 100 | 5 | 9.07 | 29832 | 89.38 | FALSE | FALSE |
| 1.6 | 100 | 6 | 7.74 | 30093 | 90.17 | FALSE | FALSE |
| 1.7 | 100 | 7 | 6.48 | 30544 | 91.52 | FALSE | FALSE |
| 1.8 | 100 | 5 | 6.12 | 30727 | 92.07 | FALSE | FALSE |
| 1.9 | 100 | 4 | 5.26 | 31063 | 93.07 | FALSE | FALSE |
| 2.0 | 100 | 4 | 5.01 | 31150 | 93.33 | FALSE | FALSE |

```r
knitr::kable(parameter_summary, caption = "Phase 7 selected DBSCAN settings")
```

Table 37: Phase 7 selected DBSCAN settings

| metric | value |
|---|---|
| kNN distance k | 5 |
| DBSCAN minPts grid | 40, 50, 75, 100 |
| DBSCAN eps grid | 0.3 to 2 |
| Largest cluster share cap (%) | 60 |
| DBSCAN minPts | 100 |
| Chosen eps (km) | 1 |
| Chosen non-noise clusters | 12 |
| Chosen noise pct | 15.22 |
| Chosen largest cluster share pct | 44.32 |
| Selection reason | Accepted trial balancing cluster count, noise level, and hotspot size concentration |

```r
knn_plot_data <- data.table(
  point_order = seq_along(knn_sorted),
  distance_km = knn_sorted
)

knn_plot <- ggplot2::ggplot(
  knn_plot_data,
  ggplot2::aes(x = point_order, y = distance_km)
) +
  ggplot2::geom_line(color = "#1f78b4", linewidth = 0.6) +
  ggplot2::geom_hline(yintercept = eps_chosen, linetype = "dashed", color = "#e31a1c") +
  ggplot2::labs(
    title = "Phase 7 kNN Distance Plot",
    x = "Points sorted by 5-NN distance",
    y = "5-NN distance (km)"
  ) +
  ggplot2::theme_minimal()

print(knn_plot)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase7_knn_distance_plot.png"),
  plot = knn_plot,
  width = 7,
  height = 4.5,
  dpi = 300
)
```

```
db_result <- dbscan::dbscan(df_spatial, eps = eps_chosen, minPts = minPts_chosen)

df_spatial_clusters <- copy(df_spatial)
df_spatial_clusters[, spatial_cluster := as.integer(db_result$cluster)]

spatial_cluster_sizes <- df_spatial_clusters[, .N, by = spatial_cluster][order(spatial_cluster)]
largest_non_noise_size <- max(spatial_cluster_sizes[spatial_cluster > 0, N], na.rm = TRUE)
largest_non_noise_share_pct <- round(100 * largest_non_noise_size / nrow(df_spatial_clusters), 2)
solution_summary <- data.table(
  metric = c(
    "Rows clustered",
    "Noise pct",
    "Non-noise clusters",
    "Largest cluster size",
    "Largest cluster share pct",
    "Meets SRS cluster-count target",
    "Meets SRS noise target",
    "Meets hotspot-balance target"
  ),
  value = c(
```

```r
    nrow(df_spatial_clusters),
    round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2),
    uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]),
    largest_non_noise_size,
    largest_non_noise_share_pct,
    uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) >= 3 &
      uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) <= 15,
    round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) >= 5 &
      round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) <= 30,
    largest_non_noise_share_pct <= phase7_max_cluster_share_cap
  )
)

if (!any(df_spatial_clusters$spatial_cluster > 0)) {
  stop("Phase 7 DBSCAN produced only noise points; adjust eps or minPts.")
}

fwrite(spatial_cluster_sizes, file.path(output_dir, "phase7_dbscan_cluster_sizes.csv"))
fwrite(solution_summary, file.path(output_dir, "phase7_solution_summary.csv"))

knitr::kable(spatial_cluster_sizes, caption = "Phase 7 spatial cluster sizes")
```

Table 38: Phase 7 spatial cluster sizes

| spatial_cluster | N |
|---:|---:|
| 0 | 5079 |
| 1 | 1097 |
| 2 | 14792 |
| 3 | 11080 |
| 4 | 150 |
| 5 | 187 |
| 6 | 239 |
| 7 | 132 |
| 8 | 100 |
| 9 | 130 |
| 10 | 90 |
| 11 | 140 |
| 12 | 159 |

```r
knitr::kable(solution_summary, caption = "Phase 7 DBSCAN solution summary")
```

Table 39: Phase 7 DBSCAN solution summary

| metric | value |
|---|---:|
| Rows clustered | 33375.00 |
| Noise pct | 15.22 |
| Non-noise clusters | 12.00 |
| Largest cluster size | 14792.00 |
| Largest cluster share pct | 44.32 |
| Meets SRS cluster-count target | 1.00 |

| metric | value |
|---|---|
| Meets SRS noise target | 1.00 |
| Meets hotspot-balance target | 1.00 |

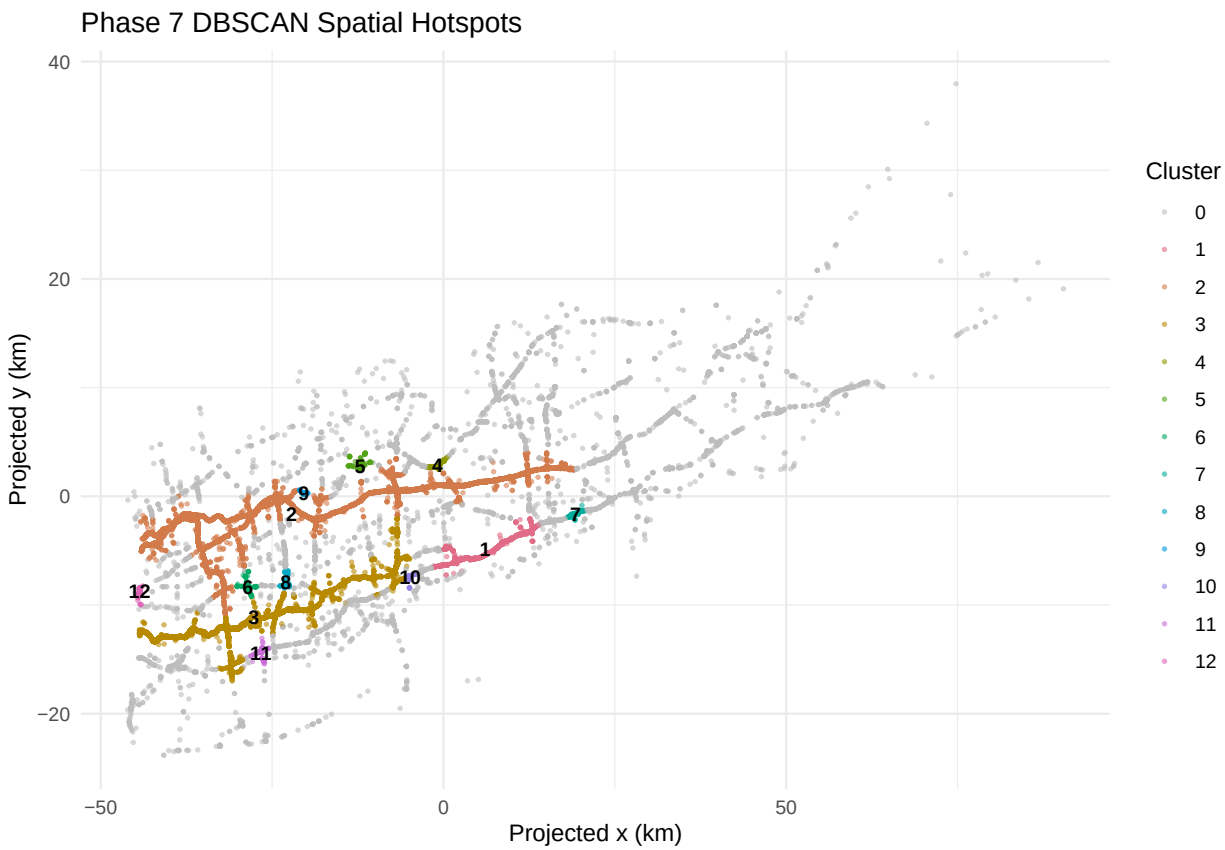
```
lat0 <- 40.8
lng0 <- -73.2

non_noise_assignments <- df_spatial_clusters[spatial_cluster > 0, .(
  cluster_size = .N,
  centroid_x_km = mean(x_km),
  centroid_y_km = mean(y_km),
  x_span_km = diff(range(x_km)),
  y_span_km = diff(range(y_km))
), by = spatial_cluster][
  , `:=`(
    centroid_lat = round((centroid_y_km / 110.574) + lat0, 5),
    centroid_lng = round((centroid_x_km / (111.32 * cos(lat0 * pi / 180))) + lng0, 5),
    shape_hint = fifelse(x_span_km >= y_span_km, "east_west", "north_south")
  )
]

non_noise_assignments[, corridor_label := dplyr::case_when(
  centroid_lat >= 40.74 & centroid_lat <= 40.76 ~ "Long Island Expressway (I-495)",
  centroid_lat >= 40.68 & centroid_lat <= 40.72 ~ "Southern State Parkway",
  centroid_lat >= 40.78 & centroid_lat <= 40.82 ~ "Northern State Parkway",
  centroid_lat >= 40.64 & centroid_lat <= 40.68 ~ "Sunrise Highway",
  centroid_lat >= 40.72 & centroid_lat <= 40.76 & shape_hint == "north_south" ~ "Meadowbrook / Wantagh P
  TRUE ~ "Surface Road Cluster"
)]

fwrite(non_noise_assignments, file.path(output_dir, "phase7_corridor_assignments.csv"))

knitr::kable(non_noise_assignments, caption = "Phase 7 corridor assignment summary")
```

Table 40: Phase 7 corridor assignment summary

| spatial_cluster | cluster_size | centroid_x_km | centroid_y_km | x_span_km | y_span_km | centroid_lat | centroid_lng | shape_hint | corridor_label |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1097 | 6.1176454 | -4.8680302 | 15.2086654 | 17.0550840 | 40.75597 | -73.12740 | east_west | Long Island Expressway (I-495) |
| 7 | 132 | 19.2660169 | -1.6064751 | 2.3867421 | 1.3434744 | 40.78547 | -72.97137 | east_west | Northern State Parkway |
| 2 | 14792 | -22.1567460 | -1.6256442 | 63.1185968 | 18.8096184 | 40.78530 | -73.46293 | east_west | Northern State Parkway |
| 3 | 11080 | -27.6724890 | -11.0638496 | 39.4362294 | 14.9291486 | 40.69994 | -73.52838 | east_west | Southern State Parkway |
| 4 | 150 | -0.8362332 | 2.9054348 | 2.9325504 | 1.0944205 | 40.82628 | -73.20992 | east_west | Surface Road Cluster |
| 5 | 187 | -12.1551395 | 2.8124230 | 3.2742599 | 1.4906484 | 40.82543 | -73.34424 | east_west | Surface Road Cluster |

| spatial_cluster | cluster_size | centroid_x_km | centroid_y_km | x_km | span_y_km | span_km | centroid_lat | centroid_lng | shape_hint | corridor_label |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | 90 | -4.8371610 | -7.4186407 | 1.5632685 | 1.3655889 | 0.73291 | -73.25740 | east_west | | Surface Road Cluster |
| 6 | 239 | -28.5097992 | -8.3009745 | 2.9317077 | 2.2750601 | 0.72493 | -73.53832 | east_west | | Surface Road Cluster |
| 8 | 100 | -22.9811129 | -7.9196066 | 1.3350688 | 1.6167025 | 0.72838 | -73.47271 | north_south | | Meadowbrook / Wantagh Pkwy |
| 9 | 130 | -20.3708466 | 0.2808053 | 1.2575417 | 0.3435534 | 0.80254 | -73.44174 | east_west | | Northern State Parkway |
| 11 | 140 | -26.6225165 | -14.4221418 | 2.7089856 | 2.3794419 | 0.66957 | -73.51592 | east_west | | Sunrise Highway |
| 12 | 159 | -44.3074014 | -8.7347346 | 0.8249062 | 1.7416511 | 0.72101 | -73.72579 | north_south | | Meadowbrook / Wantagh Pkwy |

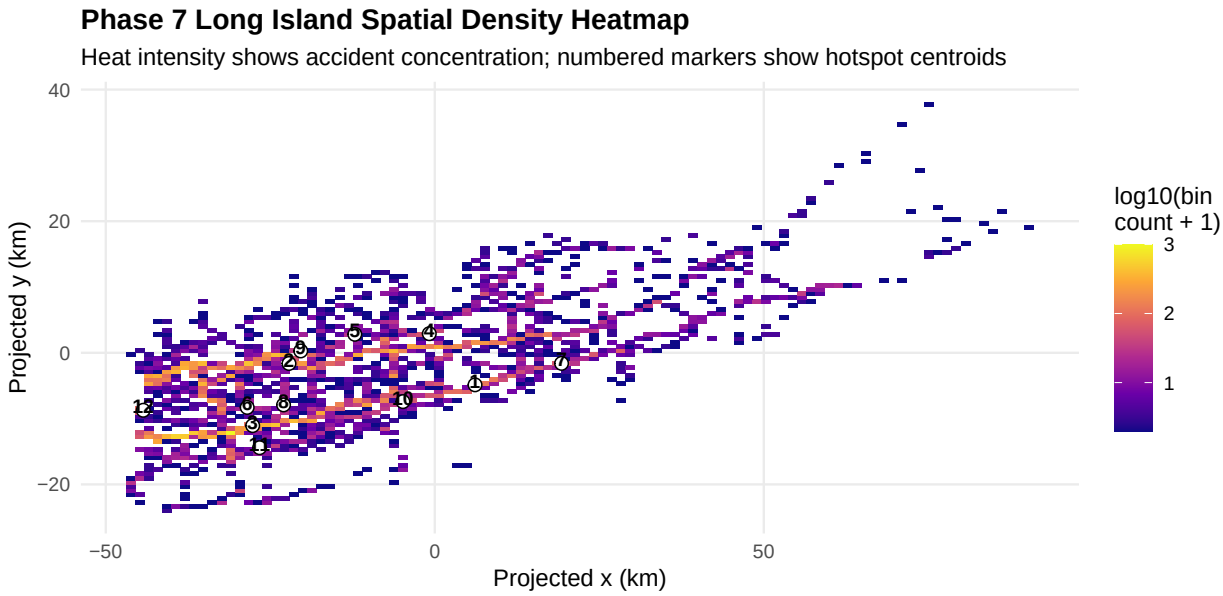
```
df_spatial_plot <- copy(df_spatial_clusters)
df_spatial_plot[, cluster_label := factor(spatial_cluster)]

cluster_levels <- sort(unique(df_spatial_plot$cluster_label))
non_noise_levels <- setdiff(as.character(cluster_levels), "0")
cluster_colors <- c(
  "0" = "#bdbdbd",
  setNames(grDevices::hcl.colors(length(non_noise_levels), palette = "Dark 3"), non_noise_levels)
)

spatial_map <- ggplot2::ggplot(
  df_spatial_plot,
  ggplot2::aes(x = x_km, y = y_km, color = cluster_label)
) +
  ggplot2::geom_point(size = 0.5, alpha = 0.6) +
  ggplot2::geom_text(
    data = non_noise_assignments,
    ggplot2::aes(x = centroid_x_km, y = centroid_y_km, label = spatial_cluster),
    inherit.aes = FALSE,
    size = 3.2,
    fontface = "bold",
    color = "black"
  ) +
  ggplot2::scale_color_manual(values = cluster_colors) +
  ggplot2::labs(
    title = "Phase 7 DBSCAN Spatial Hotspots",
    x = "Projected x (km)",
    y = "Projected y (km)",
    color = "Cluster"
  ) +
  ggplot2::theme_minimal()

print(spatial_map)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase7_spatial_hotspot_map.png"),
  plot = spatial_map,
  width = 8,
  height = 5.5,
  dpi = 300
)
```

```r
spatial_density_heatmap <- ggplot2::ggplot(
  df_spatial_plot,
  ggplot2::aes(
    x = x_km,
    y = y_km,
    fill = after_stat(log10(count + 1))
  )
) +
  ggplot2::geom_bin2d(bins = 100) +
  ggplot2::geom_point(
    data = non_noise_assignments,
    ggplot2::aes(x = centroid_x_km, y = centroid_y_km),
    inherit.aes = FALSE,
    shape = 21,
    size = 2.6,
    stroke = 0.5,
    fill = "white",
    color = "black"
```

```r
  ) +
  ggplot2::geom_text(
    data = non_noise_assignments,
    ggplot2::aes(x = centroid_x_km, y = centroid_y_km, label = spatial_cluster),
    inherit.aes = FALSE,
    size = 3,
    fontface = "bold",
    nudge_y = 0.6
  ) +
  ggplot2::scale_fill_viridis_c(
    option = "C",
    name = "log10(bin\ncount + 1)"
  ) +
  ggplot2::coord_equal() +
  ggplot2::labs(
    title = "Phase 7 Long Island Spatial Density Heatmap",
    subtitle = "Heat intensity shows accident concentration; numbered markers show hotspot centroids",
    x = "Projected x (km)",
    y = "Projected y (km)"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank()
  )

print(spatial_density_heatmap)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase7_spatial_density_heatmap.png"),
  plot = spatial_density_heatmap,
  width = 8,
  height = 5.5,
  dpi = 300
)
```

```
phase_7_summary <- data.table(
  checkpoint = c(
    "kNN distance plot saved",
    "eps trials saved",
    "DBSCAN solution created",
    "Corridor assignments saved",
    "Spatial hotspot map saved",
    "Spatial density heatmap saved"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase7_knn_distance_plot.png")),
    file.exists(file.path(output_dir, "phase7_eps_trials.csv")),
    exists("db_result"),
    file.exists(file.path(output_dir, "phase7_corridor_assignments.csv")),
    file.exists(file.path(output_dir, "phase7_spatial_hotspot_map.png")),
    file.exists(file.path(output_dir, "phase7_spatial_density_heatmap.png"))
  )
)
```

```r
fwrite(phase_7_summary, file.path(output_dir, "phase7_summary.csv"))

knitr::kable(phase_7_summary, caption = "Phase 7 readiness summary")
```

Table 41: Phase 7 readiness summary

| checkpoint | status |
|---|---|
| kNN distance plot saved | TRUE |
| eps trials saved | TRUE |
| DBSCAN solution created | TRUE |
| Corridor assignments saved | TRUE |
| Spatial hotspot map saved | TRUE |
| Spatial density heatmap saved | TRUE |

## Phase 8: Post-Clustering Interpretation

This phase joins the scenario and spatial cluster labels back to the working data and builds the interpretation tables used for the final report.

```r
kp_final <- selected_model
db_final <- db_result

if (length(kp_final$cluster) != nrow(df_phase3)) {
  stop("Scenario cluster labels do not match the Phase 3 data rows.")
}

if (length(db_final$cluster) != nrow(df_phase3)) {
  stop("Spatial cluster labels do not match the Phase 3 data rows.")
}

df_clustered <- copy(df_phase3)
df_clustered[, scenario_cluster := as.integer(kp_final$cluster)]
df_clustered[, spatial_cluster := as.integer(db_final$cluster)]

label_join_check <- data.table(
  metric = c(
    "Rows in clustered data",
    "Scenario labels",
    "Spatial labels",
    "Scenario label NAs",
    "Spatial label NAs"
  ),
  value = c(
    nrow(df_clustered),
    length(kp_final$cluster),
    length(db_final$cluster),
    sum(is.na(df_clustered$scenario_cluster)),
    sum(is.na(df_clustered$spatial_cluster))
  )
)
```

```r
if (any(label_join_check[metric %in% c("Scenario label NAs", "Spatial label NAs"), value > 0])) {
  stop("Phase 8 label joining produced missing cluster labels.")
}

fwrite(label_join_check, file.path(output_dir, "phase8_label_join_check.csv"))

knitr::kable(label_join_check, caption = "Phase 8 label join check")
```

Table 42: Phase 8 label join check

| metric | value |
|---|---|
| Rows in clustered data | 33375 |
| Scenario labels | 33375 |
| Spatial labels | 33375 |
| Scenario label NAs | 0 |
| Spatial label NAs | 0 |

```r
dataset_sev4_pct <- round(100 * mean(as.integer(df_clustered$Severity) == 4), 2)
dataset_sev34_pct <- round(100 * mean(as.integer(df_clustered$Severity) %in% c(3, 4)), 2)

scenario_severity_profile <- df_clustered[, .(
  n = .N,
  sev1_pct = round(100 * mean(as.integer(Severity) == 1), 2),
  sev2_pct = round(100 * mean(as.integer(Severity) == 2), 2),
  sev3_pct = round(100 * mean(as.integer(Severity) == 3), 2),
  sev4_pct = round(100 * mean(as.integer(Severity) == 4), 2),
  sev34_pct = round(100 * mean(as.integer(Severity) %in% c(3, 4)), 2)
), by = scenario_cluster][order(scenario_cluster)]

scenario_severity_profile[, high_risk_archetype := (
  sev4_pct >= (1.5 * dataset_sev4_pct) |
    sev34_pct == max(sev34_pct)
)]

fwrite(scenario_severity_profile, file.path(output_dir, "phase8_scenario_severity_profile.csv"))

knitr::kable(scenario_severity_profile, caption = "Phase 8 severity profile by scenario cluster")
```

Table 43: Phase 8 severity profile by scenario cluster

| scenario_cluster | n | sev1_pct | sev2_pct | sev3_pct | sev4_pct | sev34_pct | high_risk_archetype |
|---|---|---|---|---|---|---|---|
| 1 | 10389 | 0.72 | 85.29 | 12.49 | 1.49 | 13.99 | FALSE |
| 2 | 3857 | 0.34 | 82.40 | 13.20 | 4.07 | 17.27 | FALSE |
| 3 | 10509 | 0.40 | 87.72 | 9.92 | 1.97 | 11.89 | FALSE |
| 4 | 6301 | 0.43 | 82.64 | 12.71 | 4.22 | 16.93 | TRUE |
| 5 | 2319 | 0.17 | 81.63 | 12.42 | 5.78 | 18.20 | TRUE |

```r
scenario_county_profile <- df_clustered[, .(
  n = .N,
```

```
  Nassau_pct = round(100 * mean(County == "Nassau"), 2),
  Suffolk_pct = round(100 * mean(County == "Suffolk"), 2),
  Queens_pct = round(100 * mean(County == "Queens"), 2),
  dominant_county = mode_value(County)
), by = scenario_cluster][order(scenario_cluster)]

fwrite(scenario_county_profile, file.path(output_dir, "phase8_scenario_county_profile.csv"))

knitr::kable(scenario_county_profile, caption = "Phase 8 county composition by scenario cluster")
```

Table 44: Phase 8 county composition by scenario cluster

| scenario_cluster | n | Nassau_pct | Suffolk_pct | Queens_pct | dominant_county |
|---|---|---|---|---|---|
| 1 | 10389 | 53.62 | 45.87 | 0.51 | Nassau |
| 2 | 3857 | 66.74 | 32.25 | 1.01 | Nassau |
| 3 | 10509 | 57.46 | 42.34 | 0.20 | Nassau |
| 4 | 6301 | 59.80 | 39.80 | 0.40 | Nassau |
| 5 | 2319 | 62.05 | 37.56 | 0.39 | Nassau |

```
scenario_weather_time_profile <- df_clustered[, .(
  n = .N,
  dominant_weather_group = mode_value(weather_group),
  dominant_weather_share = round(100 * dominant_share(weather_group), 2),
  dominant_hour_band = mode_value(hour_band),
  dominant_hour_share = round(100 * dominant_share(hour_band), 2),
  night_pct = round(100 * mean(Sunrise_Sunset == "Night"), 2),
  weekend_pct = round(100 * mean(weekend), 2),
  mean_temperature = round(mean(`Temperature(F)`), 2),
  mean_visibility = round(mean(`Visibility(mi)`), 2),
  mean_wind_speed = round(mean(`Wind_Speed(mph)`), 2),
  junction_pct = round(100 * mean(Junction == "TRUE"), 2),
  traffic_signal_pct = round(100 * mean(Traffic_Signal == "TRUE"), 2),
  crossing_pct = round(100 * mean(Crossing == "TRUE"), 2),
  stop_pct = round(100 * mean(Stop == "TRUE"), 2)
), by = scenario_cluster][order(scenario_cluster)]

fwrite(scenario_weather_time_profile, file.path(output_dir, "phase8_scenario_weather_time_profile.csv"))

knitr::kable(scenario_weather_time_profile, caption = "Phase 8 weather and time profile by scenario clus
```

Table 45: Phase 8 weather and time profile by scenario cluster

| scenario_cluster | n | dominant_weather_group | dominant_weather_share | dominant_hour_band | dominant_hour_share | night_pct | weekend_pct | mean_temperature | mean_visibility | mean_wind_speed | junction_pct | traffic_signal_pct | crossing_pct | stop_pct |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 10389 | Cloudy | 84.45 | AM_Rush | 57.63 | 5.11 | 1.16 | 60.58 | 8.74 | 10.20 | 15.44 | 11.49 | 12.07 | 1.45 |
| 2 | 3857 | Cloudy | 75.71 | Midday | 60.18 | 11.10 | 70.52 | 59.14 | 9.09 | 12.20 | 16.90 | 7.23 | 7.49 | 0.75 |
| 3 | 10509 | Clear | 93.03 | AM_Rush | 38.69 | 8.39 | 6.05 | 60.49 | 9.67 | 9.99 | 15.65 | 8.94 | 9.45 | 0.98 |
| 4 | 6301 | Cloudy | 63.39 | Early_AM | 29.50 | 71.10 | 0.00 | 50.03 | 8.44 | 10.00 | 16.82 | 9.74 | 10.30 | 0.83 |
| 5 | 2319 | Clear | 68.56 | Early_AM | 35.36 | 90.94 | 74.39 | 57.91 | 9.38 | 7.72 | 16.00 | 6.86 | 8.15 | 0.52 |

```r
scenario_name_lookup <- copy(cluster_profile_summary)[
  , .(
    scenario_cluster = as.integer(as.character(cluster)),
    working_name
  )
]

spatial_scenario_crosstab <- df_clustered[spatial_cluster > 0, .N, by = .(spatial_cluster, scenario_clus
  order(spatial_cluster, -N, scenario_cluster)
][
  , spatial_cluster_pct := round(100 * N / sum(N), 2), by = spatial_cluster
]

spatial_scenario_crosstab <- merge(
  spatial_scenario_crosstab,
  non_noise_assignments[, .(spatial_cluster, corridor_label)],
  by = "spatial_cluster",
  all.x = TRUE
)

spatial_scenario_crosstab <- merge(
  spatial_scenario_crosstab,
  scenario_name_lookup,
  by = "scenario_cluster",
  all.x = TRUE
)

spatial_scenario_dominance <- spatial_scenario_crosstab[
  order(spatial_cluster, -N, scenario_cluster)
][
  , .SD[1], by = spatial_cluster
][
  , .(
    spatial_cluster,
    corridor_label,
    dominant_scenario_cluster = scenario_cluster,
    dominant_scenario_name = working_name,
    dominant_count = N,
    dominant_pct = spatial_cluster_pct
  )
]

fwrite(spatial_scenario_crosstab, file.path(output_dir, "phase8_spatial_scenario_crosstab.csv"))
fwrite(spatial_scenario_dominance, file.path(output_dir, "phase8_spatial_scenario_dominance.csv"))

knitr::kable(spatial_scenario_crosstab, caption = "Phase 8 spatial-scenario cross-tabulation")
```

Table 46: Phase 8 spatial-scenario cross-tabulation

| scenario_cluster | spatial_cluster | N | spatial_cluster_pct | corridor_label | working_name |
|---|---|---|---|---|---|
| 1 | 1 | 508 | 46.31 | Long Island Expressway (I-495) | Scenario Cluster 1 |

| scenario_cluster | spatial_cluster | N | spatial_cluster_pct | corridor_label | working_name |
|---|---|---|---|---|---|
| 1 | 2 | 4600 | 31.10 | Northern State Parkway | Scenario Cluster 1 |
| 1 | 3 | 3071 | 27.72 | Southern State Parkway | Scenario Cluster 1 |
| 1 | 4 | 61 | 40.67 | Surface Road Cluster | Scenario Cluster 1 |
| 1 | 5 | 57 | 30.48 | Surface Road Cluster | Scenario Cluster 1 |
| 1 | 6 | 62 | 25.94 | Surface Road Cluster | Scenario Cluster 1 |
| 1 | 7 | 69 | 52.27 | Northern State Parkway | Scenario Cluster 1 |
| 1 | 8 | 34 | 34.00 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 1 |
| 1 | 9 | 33 | 25.38 | Northern State Parkway | Scenario Cluster 1 |
| 1 | 10 | 39 | 43.33 | Surface Road Cluster | Scenario Cluster 1 |
| 1 | 11 | 34 | 24.29 | Sunrise Highway | Scenario Cluster 1 |
| 1 | 12 | 33 | 20.75 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 1 |
| 2 | 1 | 135 | 12.31 | Long Island Expressway (I-495) | Scenario Cluster 2 |
| 2 | 2 | 1809 | 12.23 | Northern State Parkway | Scenario Cluster 2 |
| 2 | 3 | 1428 | 12.89 | Southern State Parkway | Scenario Cluster 2 |
| 2 | 4 | 9 | 6.00 | Surface Road Cluster | Scenario Cluster 2 |
| 2 | 5 | 4 | 2.14 | Surface Road Cluster | Scenario Cluster 2 |
| 2 | 6 | 12 | 5.02 | Surface Road Cluster | Scenario Cluster 2 |
| 2 | 7 | 7 | 5.30 | Northern State Parkway | Scenario Cluster 2 |
| 2 | 8 | 3 | 3.00 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 2 |
| 2 | 9 | 5 | 3.85 | Northern State Parkway | Scenario Cluster 2 |
| 2 | 10 | 12 | 13.33 | Surface Road Cluster | Scenario Cluster 2 |
| 2 | 11 | 6 | 4.29 | Sunrise Highway | Scenario Cluster 2 |
| 2 | 12 | 30 | 18.87 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 2 |
| 3 | 1 | 241 | 21.97 | Long Island Expressway (I-495) | Scenario Cluster 3 |
| 3 | 2 | 4561 | 30.83 | Northern State Parkway | Scenario Cluster 3 |
| 3 | 3 | 3591 | 32.41 | Southern State Parkway | Scenario Cluster 3 |

| scenario_cluster | spatial_cluster | N | spatial_cluster_pct | corridor_label | working_name |
|---|---|---|---|---|---|
| 3 | 4 | 26 | 17.33 | Surface Road Cluster | Scenario Cluster 3 |
| 3 | 5 | 88 | 47.06 | Surface Road Cluster | Scenario Cluster 3 |
| 3 | 6 | 107 | 44.77 | Surface Road Cluster | Scenario Cluster 3 |
| 3 | 7 | 27 | 20.45 | Northern State Parkway | Scenario Cluster 3 |
| 3 | 8 | 31 | 31.00 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 3 |
| 3 | 9 | 51 | 39.23 | Northern State Parkway | Scenario Cluster 3 |
| 3 | 10 | 8 | 8.89 | Surface Road Cluster | Scenario Cluster 3 |
| 3 | 11 | 60 | 42.86 | Sunrise Highway | Scenario Cluster 3 |
| 3 | 12 | 24 | 15.09 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 3 |
| 4 | 1 | 163 | 14.86 | Long Island Expressway (I-495) | Scenario Cluster 4 |
| 4 | 2 | 2832 | 19.15 | Northern State Parkway | Scenario Cluster 4 |
| 4 | 3 | 2132 | 19.24 | Southern State Parkway | Scenario Cluster 4 |
| 4 | 4 | 45 | 30.00 | Surface Road Cluster | Scenario Cluster 4 |
| 4 | 5 | 24 | 12.83 | Surface Road Cluster | Scenario Cluster 4 |
| 4 | 6 | 36 | 15.06 | Surface Road Cluster | Scenario Cluster 4 |
| 4 | 7 | 22 | 16.67 | Northern State Parkway | Scenario Cluster 4 |
| 4 | 8 | 26 | 26.00 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 4 |
| 4 | 9 | 35 | 26.92 | Northern State Parkway | Scenario Cluster 4 |
| 4 | 10 | 22 | 24.44 | Surface Road Cluster | Scenario Cluster 4 |
| 4 | 11 | 29 | 20.71 | Sunrise Highway | Scenario Cluster 4 |
| 4 | 12 | 57 | 35.85 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 4 |
| 5 | 1 | 50 | 4.56 | Long Island Expressway (I-495) | Scenario Cluster 5 |
| 5 | 2 | 990 | 6.69 | Northern State Parkway | Scenario Cluster 5 |
| 5 | 3 | 858 | 7.74 | Southern State Parkway | Scenario Cluster 5 |
| 5 | 4 | 9 | 6.00 | Surface Road Cluster | Scenario Cluster 5 |
| 5 | 5 | 14 | 7.49 | Surface Road Cluster | Scenario Cluster 5 |

| scenario_cluster | spatial_cluster | N | spatial_cluster_pct | corridor_label | working_name |
|---|---|---|---|---|---|
| 5 | 6 | 22 | 9.21 | Surface Road Cluster | Scenario Cluster 5 |
| 5 | 7 | 7 | 5.30 | Northern State Parkway | Scenario Cluster 5 |
| 5 | 8 | 6 | 6.00 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 5 |
| 5 | 9 | 6 | 4.62 | Northern State Parkway | Scenario Cluster 5 |
| 5 | 10 | 9 | 10.00 | Surface Road Cluster | Scenario Cluster 5 |
| 5 | 11 | 11 | 7.86 | Sunrise Highway | Scenario Cluster 5 |
| 5 | 12 | 15 | 9.43 | Meadowbrook / Wantagh Pkwy | Scenario Cluster 5 |

```
knitr::kable(spatial_scenario_dominance, caption = "Phase 8 dominant scenario by spatial hotspot")
```

Table 47: Phase 8 dominant scenario by spatial hotspot

| spatial_cluster | corridor_label | dominant_scenario_cluster | dominant_scenario_name | dominant_count | dominant_pct |
|---|---|---|---|---|---|
| 1 | Long Island Expressway (I-495) | 1 | Scenario Cluster 1 | 508 | 46.31 |
| 2 | Northern State Parkway | 1 | Scenario Cluster 1 | 4600 | 31.10 |
| 3 | Southern State Parkway | 3 | Scenario Cluster 3 | 3591 | 32.41 |
| 4 | Surface Road Cluster | 1 | Scenario Cluster 1 | 61 | 40.67 |
| 5 | Surface Road Cluster | 3 | Scenario Cluster 3 | 88 | 47.06 |
| 6 | Surface Road Cluster | 3 | Scenario Cluster 3 | 107 | 44.77 |
| 7 | Northern State Parkway | 1 | Scenario Cluster 1 | 69 | 52.27 |
| 8 | Meadowbrook / Wantagh Pkwy | 1 | Scenario Cluster 1 | 34 | 34.00 |
| 9 | Northern State Parkway | 3 | Scenario Cluster 3 | 51 | 39.23 |
| 10 | Surface Road Cluster | 1 | Scenario Cluster 1 | 39 | 43.33 |
| 11 | Sunrise Highway | 3 | Scenario Cluster 3 | 60 | 42.86 |
| 12 | Meadowbrook / Wantagh Pkwy | 4 | Scenario Cluster 4 | 57 | 35.85 |

```
scenario_final_names <- merge(
  scenario_weather_time_profile,
  scenario_severity_profile[, .(scenario_cluster, sev4_pct, high_risk_archetype)],
  by = "scenario_cluster",
```

```
  all.x = TRUE
)

scenario_final_names <- merge(
  scenario_final_names,
  scenario_name_lookup,
  by = "scenario_cluster",
  all.x = TRUE
)

scenario_final_names[, final_name := dplyr::case_when(
  dominant_weather_group == "Snow_Ice" ~ "Winter Adverse --- Snow and Ice",
  night_pct >= 50 & dominant_weather_group %in% c("Fog_Haze", "Rain", "Thunderstorm") ~ "Night Low-Visi
  night_pct >= 50 & dominant_hour_band == "Early_AM" ~ paste(
    "Night / Early AM ---",
    gsub("_", " ", dominant_weather_group)
  ),
  weekend_pct >= 40 & traffic_signal_pct >= 8 ~ "Weekend Surface Road --- Signal Heavy",
  dominant_hour_band %in% c("AM_Rush", "PM_Rush") & junction_pct >= 10 ~ paste(
    gsub("_", " ", dominant_weather_group),
    gsub("_", " ", dominant_hour_band),
    "--- Junction Heavy"
  ),
  TRUE ~ paste(
    "Scenario Cluster",
    scenario_cluster,
    "---",
    gsub("_", " ", dominant_weather_group)
  )
)]

scenario_final_names[, final_name := ifelse(
  high_risk_archetype,
  paste(final_name, "(High Risk)"),
  final_name
)]

duplicate_scenario_names <- scenario_final_names[, .N, by = final_name][N > 1, final_name]
if (length(duplicate_scenario_names) > 0) {
  scenario_final_names[final_name %in% duplicate_scenario_names, final_name := paste0(final_name, " (S"
}

spatial_final_names <- rbind(
  data.table(
    spatial_cluster = 0L,
    corridor_label = "Noise / Unassigned",
    final_name = "Noise / Unassigned"
  ),
  non_noise_assignments[, .(
    spatial_cluster,
    corridor_label,
    final_name = ifelse(
      corridor_label == "Surface Road Cluster",
```

```r
      paste("Surface Road Cluster", spatial_cluster),
      paste(corridor_label, "Cluster", spatial_cluster)
    )
  )]
)

scenario_name_map <- scenario_final_names$final_name
names(scenario_name_map) <- scenario_final_names$scenario_cluster

spatial_name_map <- spatial_final_names$final_name
names(spatial_name_map) <- spatial_final_names$spatial_cluster

df_clustered[, scenario_cluster_name := scenario_name_map[as.character(scenario_cluster)]]
df_clustered[, spatial_cluster_name := spatial_name_map[as.character(spatial_cluster)]]

if ("working_name" %in% names(spatial_scenario_crosstab)) {
  spatial_scenario_crosstab[, working_name := NULL]
}
spatial_scenario_crosstab <- merge(
  spatial_scenario_crosstab,
  scenario_final_names[, .(scenario_cluster, final_name)],
  by = "scenario_cluster",
  all.x = TRUE
)

spatial_scenario_dominance <- merge(
  spatial_scenario_dominance[, !c("dominant_scenario_name"), with = FALSE],
  scenario_final_names[, .(scenario_cluster, final_name)],
  by.x = "dominant_scenario_cluster",
  by.y = "scenario_cluster",
  all.x = TRUE
)
data.table::setnames(spatial_scenario_dominance, "final_name", "dominant_scenario_name")

fwrite(spatial_scenario_crosstab, file.path(output_dir, "phase8_spatial_scenario_crosstab.csv"))
fwrite(spatial_scenario_dominance, file.path(output_dir, "phase8_spatial_scenario_dominance.csv"))

fwrite(scenario_final_names, file.path(output_dir, "phase8_scenario_final_names.csv"))
fwrite(spatial_final_names, file.path(output_dir, "phase8_spatial_final_names.csv"))

knitr::kable(scenario_final_names, caption = "Phase 8 final scenario cluster names")
```

Table 48: Phase 8 final scenario cluster names

| scenario_cluster | dominant_weather | dominant_time_period | dominant_day_type | dominant_temperature_cat | dominant_visibility_cat | dominant_speed_cat | dominant_light_cat | dominant_road_type | dominant_surface_cat | dominant_lighting_cat | dominant_weather_cat | dominant_speed_cat | dominant_light_cat | dominant_road_type | dominant_surface_cat | dominant_lighting_cat | working_name | final_name |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1038 | Cloudy | 84.45 | AM_Rush | 57.63 | 5.11 | 1.16 | 60.58 | 8.74 | 10.20 | 15.44 | 11.49 | 12.07 | 1.45 | 1.49 | FALSE | Scenario Cluster 1 | Cloudy AM Rush — Junction Heavy |

| scenario_cluster | dominant_weather | weather_mix_index | time_bin | mean_speed | mean_delay | mean_lat | mean_lon | mean_temp | mean_precip | mean_visibility | mean_wind | mean_pressure | mean_humidity | mean_hour | mean_dow | mean_month | is_peak | is_night | is_weekend | high_risk_rate | n_crashes | final_name |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 3857 | Cloudy | 75.71 | Midday | 60.18 | 11.10 | 70.52 | 59.14 | 9.09 | 12.20 | 16.90 | 7.23 | 7.49 | 0.75 | 4.07 | FALSE | Scenario Cluster 2 | Scenario Cluster 2 — Cloudy |
| 3 | 10509 | Clear | 93.03 | AM_Rush | 38.69 | 8.39 | 6.05 | 60.49 | 9.67 | 9.99 | 15.65 | 8.94 | 9.45 | 0.98 | 1.97 | FALSE | Scenario Cluster 3 | Clear AM Rush — Junction |
| 4 | 6301 | Cloudy | 63.39 | Early_AM | 29.50 | 71.10 | 0.00 | 50.03 | 8.44 | 10.00 | 16.82 | 9.74 | 10.30 | 0.83 | 4.22 | TRUE | Scenario Cluster 4 | Heavy Night / Early AM — Cloudy (High Risk) |
| 5 | 2319 | Clear | 68.56 | Early_AM | 33.36 | 90.94 | 74.39 | 57.91 | 9.38 | 7.72 | 16.00 | 6.86 | 8.15 | 0.52 | 5.78 | TRUE | Scenario Cluster 5 | Night / Early AM — Clear (High Risk) |

```
knitr::kable(spatial_final_names, caption = "Phase 8 final spatial cluster names")
```

Table 49: Phase 8 final spatial cluster names

| spatial_cluster | corridor_label | final_name |
|---|---|---|
| 0 | Noise / Unassigned | Noise / Unassigned |
| 1 | Long Island Expressway (I-495) | Long Island Expressway (I-495) Cluster 1 |
| 7 | Northern State Parkway | Northern State Parkway Cluster 7 |
| 2 | Northern State Parkway | Northern State Parkway Cluster 2 |
| 3 | Southern State Parkway | Southern State Parkway Cluster 3 |
| 4 | Surface Road Cluster | Surface Road Cluster 4 |
| 5 | Surface Road Cluster | Surface Road Cluster 5 |
| 10 | Surface Road Cluster | Surface Road Cluster 10 |
| 6 | Surface Road Cluster | Surface Road Cluster 6 |
| 8 | Meadowbrook / Wantagh Pkwy | Meadowbrook / Wantagh Pkwy Cluster 8 |
| 9 | Northern State Parkway | Northern State Parkway Cluster 9 |
| 11 | Sunrise Highway | Sunrise Highway Cluster 11 |
| 12 | Meadowbrook / Wantagh Pkwy | Meadowbrook / Wantagh Pkwy Cluster 12 |

```
phase_8_summary <- data.table(
  checkpoint = c(
    "Label join saved",
```

```r
    "Severity profile saved",
    "County profile saved",
    "Weather-time profile saved",
    "Spatial-scenario cross-tab saved",
    "Final cluster names saved",
    "Scenario names are unique"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase8_label_join_check.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_severity_profile.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_county_profile.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_weather_time_profile.csv")),
    file.exists(file.path(output_dir, "phase8_spatial_scenario_crosstab.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_final_names.csv")) &
      file.exists(file.path(output_dir, "phase8_spatial_final_names.csv")),
    uniqueN(scenario_final_names$final_name) == nrow(scenario_final_names)
  )
)

fwrite(phase_8_summary, file.path(output_dir, "phase8_summary.csv"))

knitr::kable(phase_8_summary, caption = "Phase 8 readiness summary")
```

Table 50: Phase 8 readiness summary

| checkpoint | status |
|---|---|
| Label join saved | TRUE |
| Severity profile saved | TRUE |
| County profile saved | TRUE |
| Weather-time profile saved | TRUE |
| Spatial-scenario cross-tab saved | TRUE |
| Final cluster names saved | TRUE |
| Scenario names are unique | TRUE |

## Phase 9: Export and Visualization

This phase organizes the report-ready figures, tables, and labeled dataset export so the final write-up can pull from a single output location.

```r
figure_dir <- file.path(output_dir, "figures")
table_dir <- file.path(output_dir, "tables")
interactive_dir <- file.path(output_dir, "interactive")

dir.create(figure_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(table_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(interactive_dir, recursive = TRUE, showWarnings = FALSE)

export_setup <- data.table(
  item = c("Figure directory", "Table directory", "Interactive directory"),
  path = c(figure_dir, table_dir, interactive_dir),
```

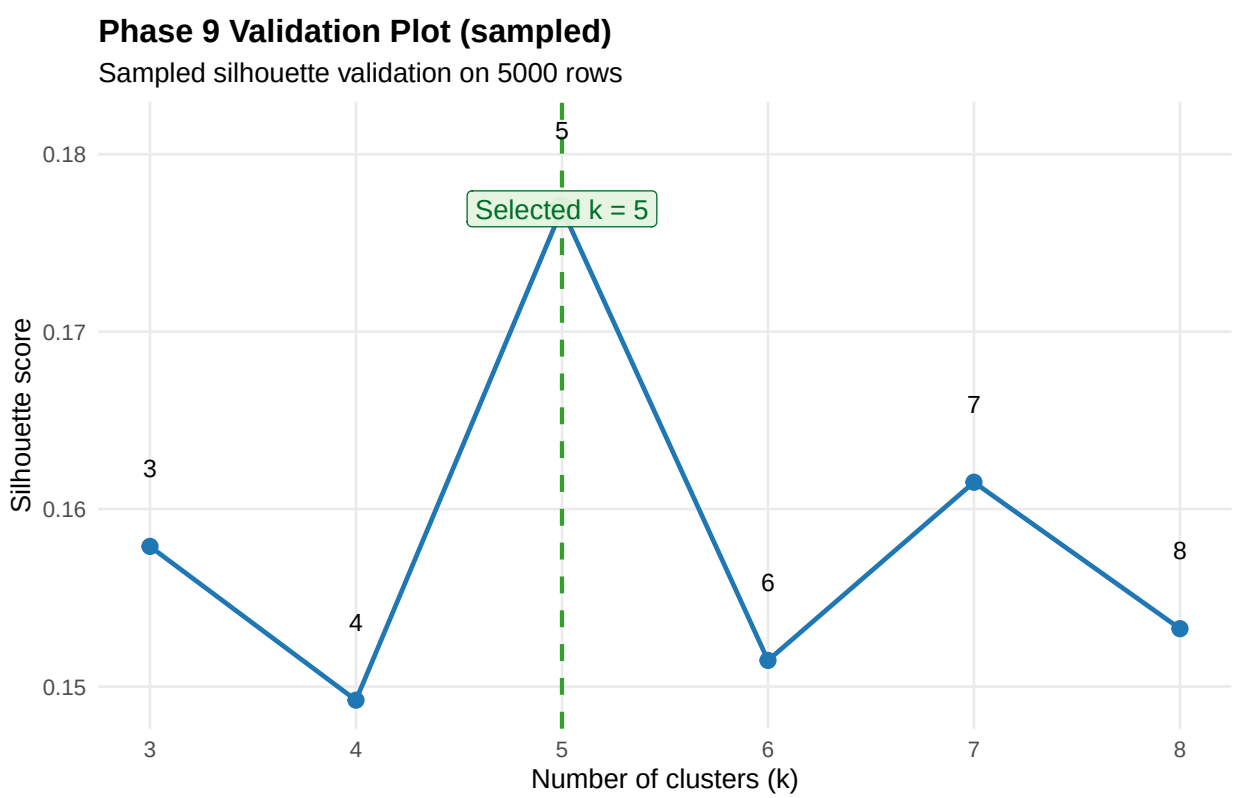
```r
  status = c(
    "Created",
    "Created",
    ifelse(interactive_exports_enabled, "Created", "Created for optional HTML exports")
  )
)

knitr::kable(export_setup, caption = "Phase 9 export folders")
```

Table 51: Phase 9 export folders

| item | path | status |
|---|---|---|
| Figure directory | local_artifacts/output/figures | Created |
| Table directory | local_artifacts/output/tables | Created |
| Interactive directory | local_artifacts/output/interactive | Created for optional HTML exports |

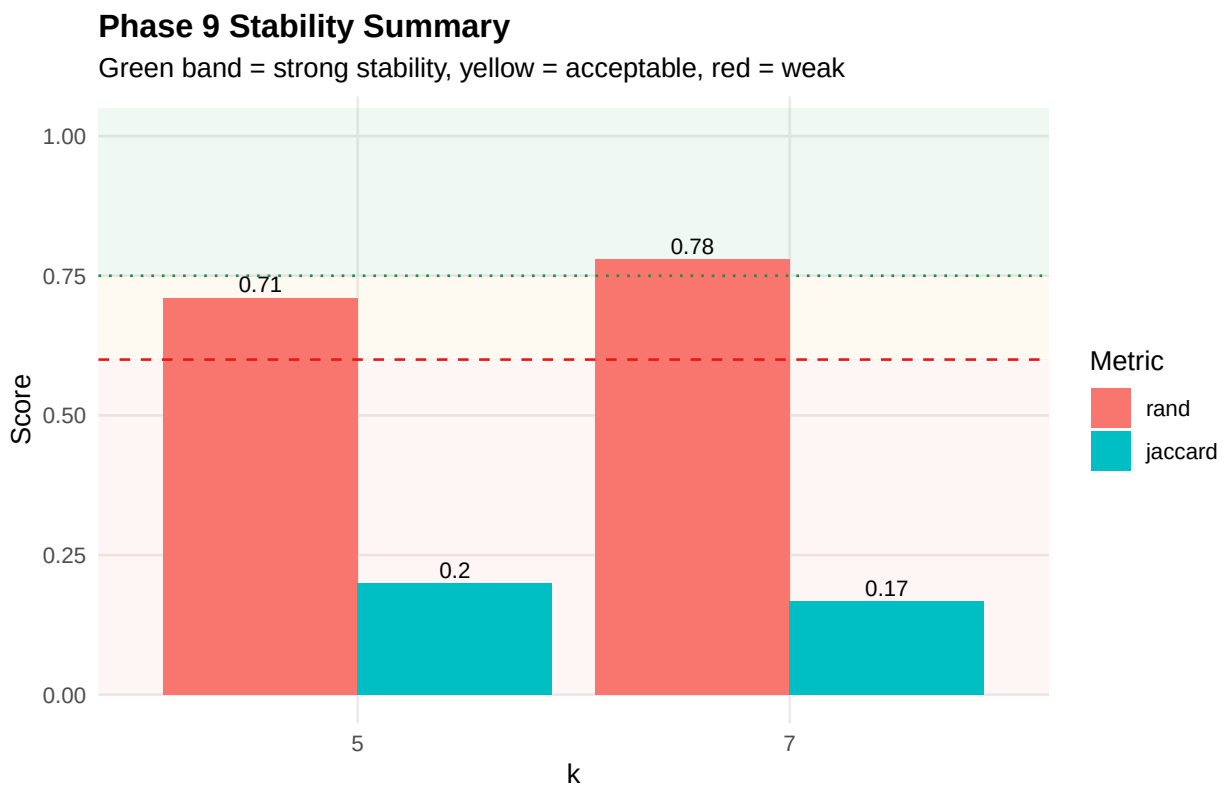
```r
selected_validation_point <- validation_table[k == selected_k]
validation_subtitle <- if (phase6_validation_mode == "consensus_sampled") {
  paste(
    "Consensus silhouette validation across",
    phase6_validation_repeats,
    "repeats of",
    validation_rows,
    "rows"
  )
} else if (phase6_validation_mode == "sampled") {
  paste("Sampled silhouette validation on", validation_rows, "rows")
} else {
  "Full-data silhouette validation"
}

validation_plot <- ggplot2::ggplot(
  validation_table,
  ggplot2::aes(x = k, y = silhouette)
) +
  ggplot2::geom_line(color = "#1f78b4", linewidth = 0.9) +
  ggplot2::geom_point(color = "#1f78b4", size = 2.6) +
  ggplot2::geom_vline(xintercept = selected_k, linetype = "dashed", color = "#33a02c", linewidth = 0.8)
  ggplot2::geom_point(
    data = selected_validation_point,
    color = "#33a02c",
    size = 4
  ) +
  ggplot2::geom_text(
    ggplot2::aes(label = k),
    nudge_y = max(validation_table$silhouette) * 0.025,
    size = 3.5
  ) +
  ggplot2::annotate(
    "label",
    x = selected_k,
    y = selected_validation_point$silhouette[[1]],
```

```r
    label = paste("Selected k =", selected_k),
    fill = "#e5f5e0",
    color = "#006d2c",
    label.size = 0.25,
    alpha = 0.95
  ) +
  ggplot2::labs(
    title = paste("Phase 9 Validation Plot (", phase6_validation_mode, ")", sep = ""),
    subtitle = validation_subtitle,
    x = "Number of clusters (k)",
    y = "Silhouette score"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank()
  )

print(validation_plot)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_validation_index_plot.png"),
  plot = validation_plot,
  width = 7,
  height = 4.5,
  dpi = 300
)
```

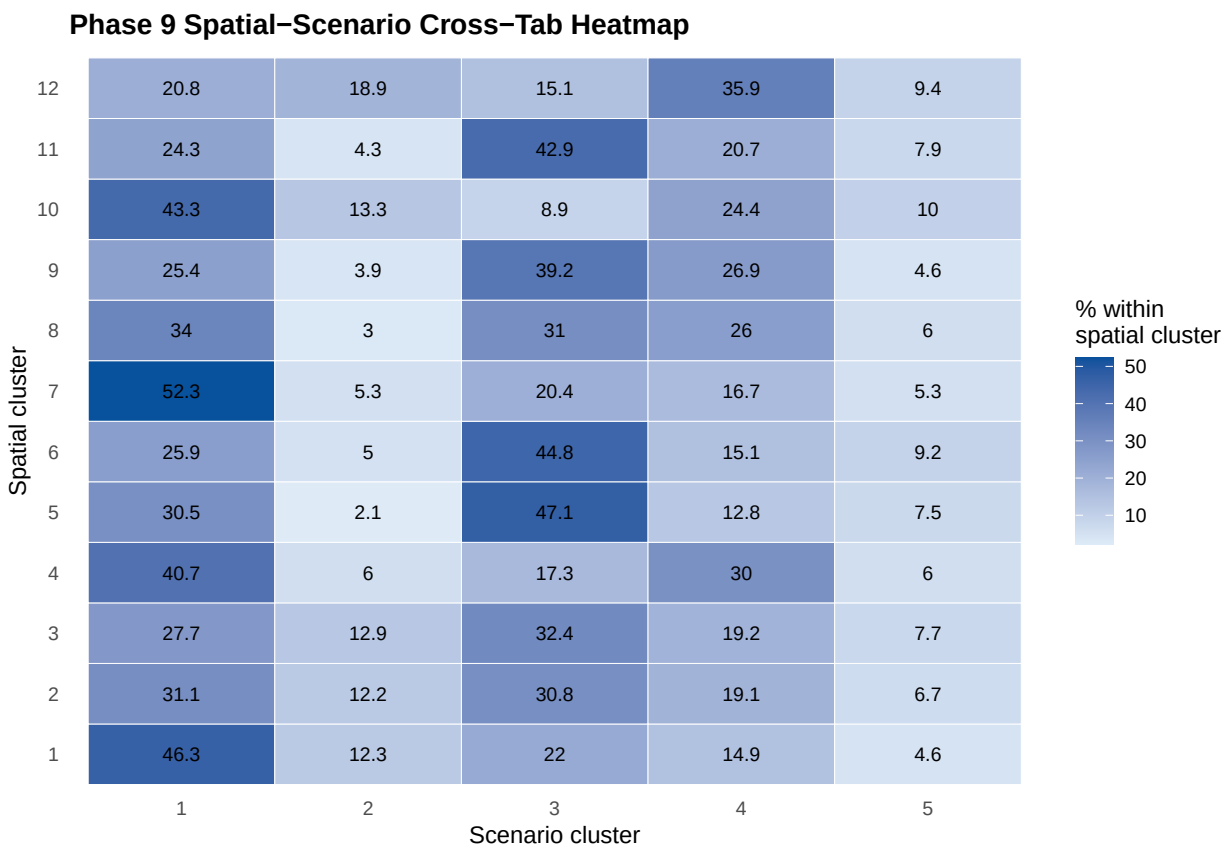
```r
stability_plot_data <- data.table::melt(
  copy(stability_summary),
  id.vars = "k",
  measure.vars = c("rand", "jaccard"),
  variable.name = "metric",
  value.name = "score"
)

stability_plot <- ggplot2::ggplot(
  stability_plot_data,
  ggplot2::aes(x = factor(k), y = score, fill = metric)
) +
  ggplot2::annotate("rect", xmin = -Inf, xmax = Inf, ymin = 0.75, ymax = 1.05, alpha = 0.08, fill = "#3
  ggplot2::annotate("rect", xmin = -Inf, xmax = Inf, ymin = 0.60, ymax = 0.75, alpha = 0.08, fill = "#f
  ggplot2::annotate("rect", xmin = -Inf, xmax = Inf, ymin = 0.00, ymax = 0.60, alpha = 0.06, fill = "#f
  ggplot2::geom_col(position = "dodge") +
  ggplot2::geom_text(
    ggplot2::aes(label = round(score, 2)),
    position = ggplot2::position_dodge(width = 0.9),
    vjust = -0.35,
    size = 3.2
  ) +
  ggplot2::geom_hline(yintercept = 0.6, linetype = "dashed", color = "#e31a1c") +
  ggplot2::geom_hline(yintercept = 0.75, linetype = "dotted", color = "#238b45") +
  ggplot2::labs(
    title = "Phase 9 Stability Summary",
    subtitle = if (phase6_stability_mode == "consensus_sampled") {
      paste(
        "Green = strong, yellow = acceptable, red = weak;",
        phase6_stability_repeats,
        "repeats with",
        phase6_stability_bootstraps,
        "bootstraps each"
      )
    } else {
      "Green band = strong stability, yellow = acceptable, red = weak"
    },
    x = "k",
    y = "Score",
    fill = "Metric"
  ) +
  ggplot2::coord_cartesian(ylim = c(0, 1.02)) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank()
  )

print(stability_plot)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_stability_plot.png"),
  plot = stability_plot,
  width = 7,
  height = 4.5,
  dpi = 300
)
```

```
heatmap_data <- merge(
  spatial_scenario_crosstab,
  scenario_final_names[, .(scenario_cluster, scenario_name_short = wrap_label(final_name, width = 18))]
  by = "scenario_cluster",
  all.x = TRUE
)

heatmap_data <- merge(
  heatmap_data,
  spatial_final_names[, .(spatial_cluster, spatial_name_short = wrap_label(final_name, width = 20))],
  by = "spatial_cluster",
  all.x = TRUE
)

heatmap_plot <- ggplot2::ggplot(
  heatmap_data,
  ggplot2::aes(
    x = factor(scenario_cluster),
    y = factor(spatial_cluster),
    fill = spatial_cluster_pct
```

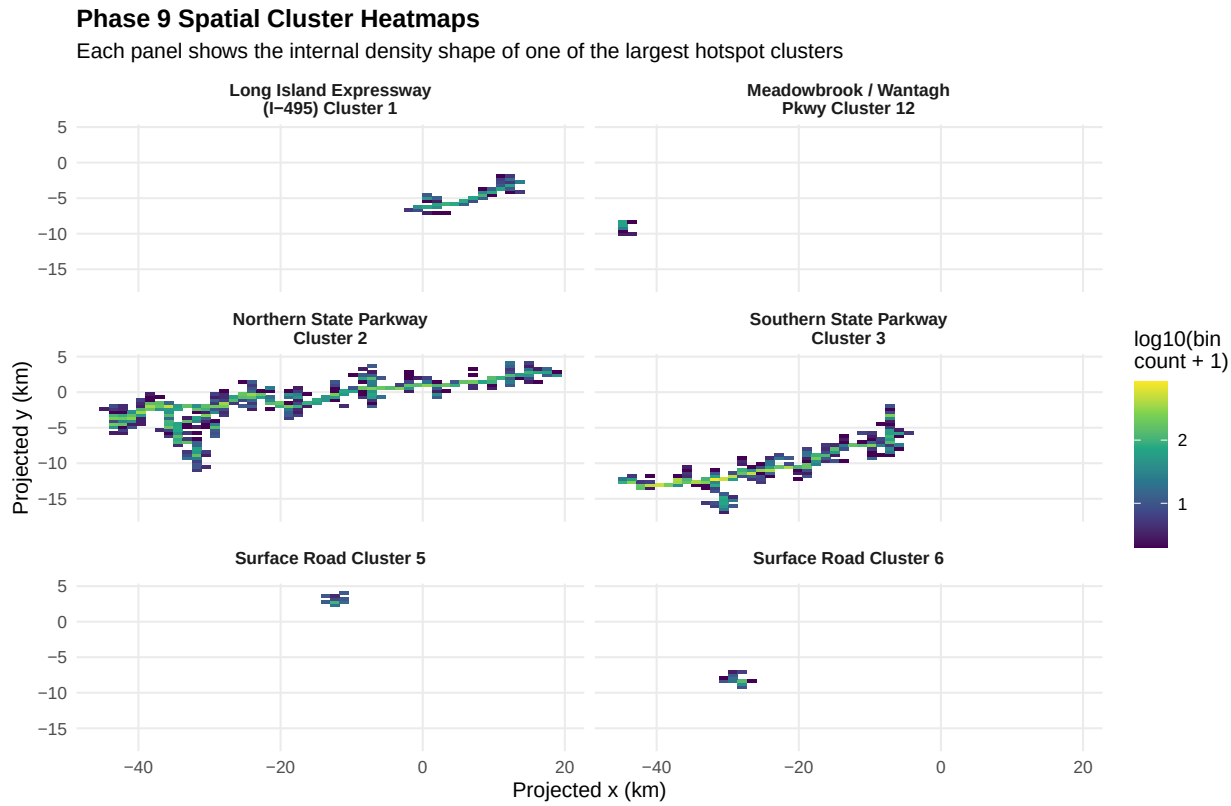
```r
  )
) +
  ggplot2::geom_tile(color = "white") +
  ggplot2::geom_text(ggplot2::aes(label = round(spatial_cluster_pct, 1)), size = 3) +
  ggplot2::scale_fill_gradient(low = "#deebf7", high = "#08519c") +
  ggplot2::labs(
    title = "Phase 9 Spatial-Scenario Cross-Tab Heatmap",
    x = "Scenario cluster",
    y = "Spatial cluster",
    fill = "% within\nspatial cluster"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid = ggplot2::element_blank()
  )

print(heatmap_plot)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_spatial_scenario_heatmap.png"),
  plot = heatmap_plot,
  width = 8,
  height = 5.5,
  dpi = 300
```

```
)
```

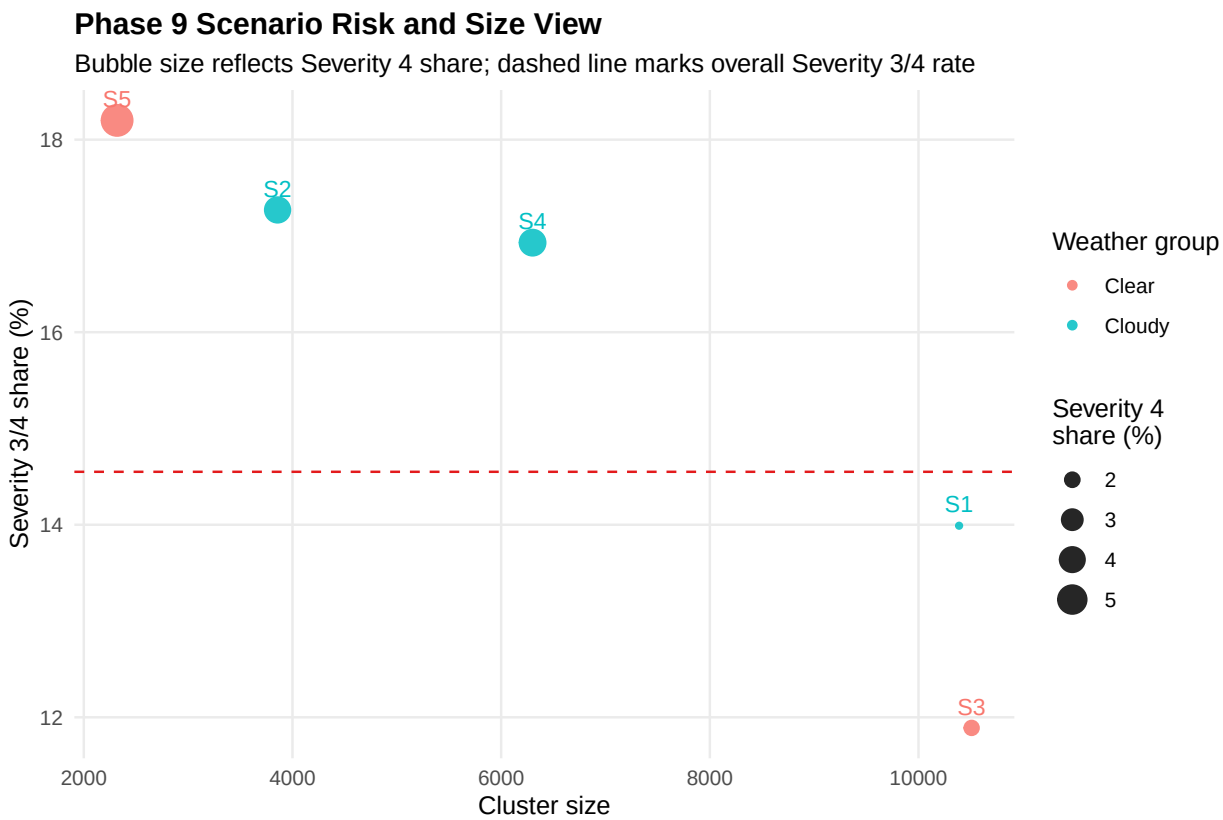
```
top_spatial_clusters <- non_noise_assignments[order(-cluster_size)][1:min(.N, 6), spatial_cluster]

spatial_cluster_heatmap_data <- merge(
  df_spatial_plot[spatial_cluster %in% top_spatial_clusters],
  spatial_final_names[, .(spatial_cluster, final_name)],
  by = "spatial_cluster",
  all.x = TRUE
)

spatial_cluster_heatmap_data[, final_name_short := wrap_label(final_name, width = 26)]

spatial_cluster_facet_heatmap <- ggplot2::ggplot(
  spatial_cluster_heatmap_data,
  ggplot2::aes(
    x = x_km,
    y = y_km,
    fill = after_stat(log10(count + 1))
  )
) +
  ggplot2::geom_bin2d(bins = 50) +
  ggplot2::scale_fill_viridis_c(
    option = "D",
    name = "log10(bin\ncount + 1)"
  ) +
  ggplot2::coord_equal() +
  ggplot2::facet_wrap(~ final_name_short, ncol = 2) +
  ggplot2::labs(
    title = "Phase 9 Spatial Cluster Heatmaps",
    subtitle = "Each panel shows the internal density shape of one of the largest hotspot clusters",
    x = "Projected x (km)",
    y = "Projected y (km)"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank(),
    strip.text = ggplot2::element_text(face = "bold")
  )

print(spatial_cluster_facet_heatmap)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_spatial_cluster_facets_heatmap.png"),
  plot = spatial_cluster_facet_heatmap,
  width = 9,
  height = 7,
  dpi = 300
)
```

```
dataset_sev34_pct <- round(100 * mean(as.integer(df_clustered$Severity) %in% c(3, 4)), 2)

risk_bubble_data <- merge(
  scenario_severity_profile,
  scenario_weather_time_profile[, .(
    scenario_cluster,
    dominant_weather_group,
    dominant_hour_band,
    mean_visibility
  )],
  by = "scenario_cluster",
  all.x = TRUE
)
```

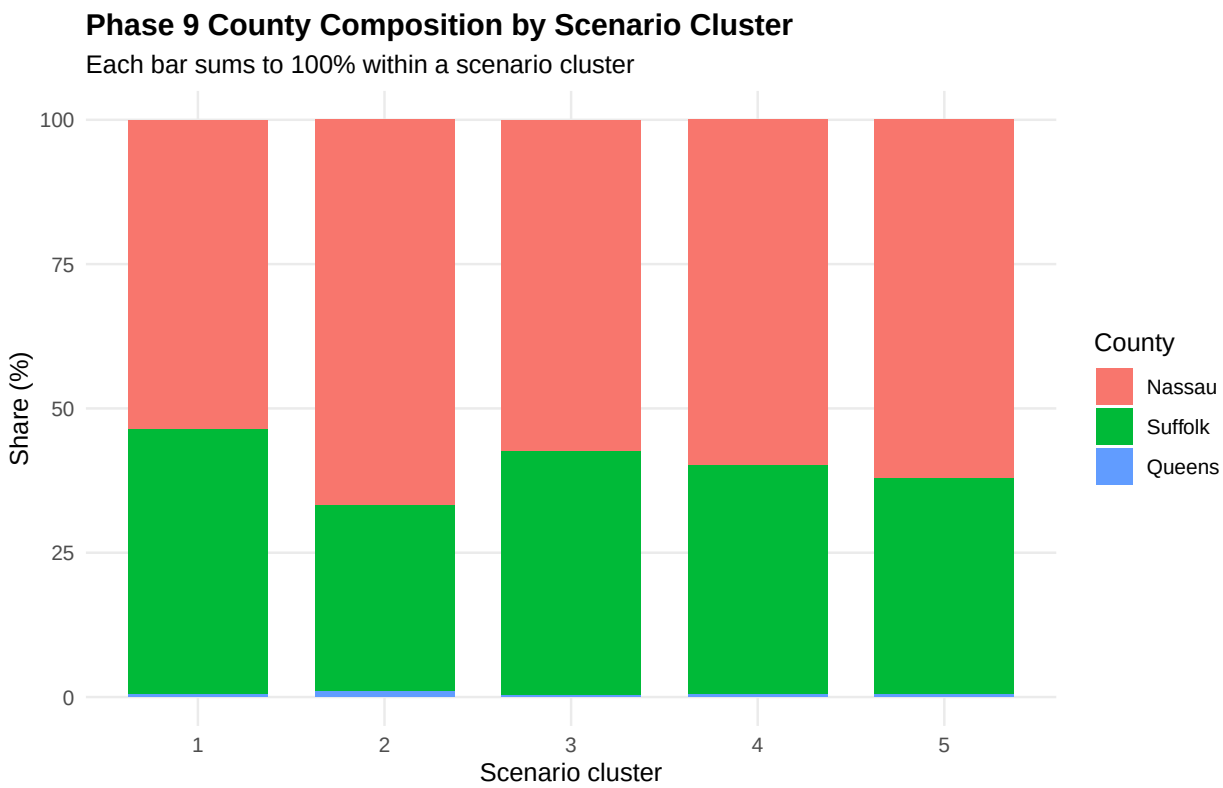
```r
risk_bubble_data <- merge(
  risk_bubble_data,
  scenario_final_names[, .(scenario_cluster, final_name)],
  by = "scenario_cluster",
  all.x = TRUE
)

risk_bubble_data[, short_label := paste0("S", scenario_cluster)]

risk_bubble_plot <- ggplot2::ggplot(
  risk_bubble_data,
  ggplot2::aes(
    x = n,
    y = sev34_pct,
    size = pmax(sev4_pct, 0.1),
    color = dominant_weather_group
  )
) +
  ggplot2::geom_hline(yintercept = dataset_sev34_pct, linetype = "dashed", color = "#e31a1c") +
  ggplot2::geom_point(alpha = 0.85) +
  ggplot2::geom_text(ggplot2::aes(label = short_label), vjust = -0.8, size = 3.4, show.legend = FALSE) +
  ggplot2::labs(
    title = "Phase 9 Scenario Risk and Size View",
    subtitle = "Bubble size reflects Severity 4 share; dashed line marks overall Severity 3/4 rate",
    x = "Cluster size",
    y = "Severity 3/4 share (%)",
    size = "Severity 4\nshare (%)",
    color = "Weather group"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank()
  )

print(risk_bubble_plot)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_scenario_risk_bubble.png"),
  plot = risk_bubble_plot,
  width = 7.5,
  height = 5,
  dpi = 300
)
```

```
county_plot_data <- data.table::melt(
  copy(scenario_county_profile),
  id.vars = c("scenario_cluster", "n", "dominant_county"),
  measure.vars = c("Nassau_pct", "Suffolk_pct", "Queens_pct"),
  variable.name = "county",
  value.name = "pct"
)

county_plot_data[, county := factor(
  county,
  levels = c("Nassau_pct", "Suffolk_pct", "Queens_pct"),
  labels = c("Nassau", "Suffolk", "Queens")
)]

county_plot <- ggplot2::ggplot(
  county_plot_data,
  ggplot2::aes(x = factor(scenario_cluster), y = pct, fill = county)
) +
  ggplot2::geom_col(width = 0.75) +
```

```r
  ggplot2::labs(
    title = "Phase 9 County Composition by Scenario Cluster",
    subtitle = "Each bar sums to 100% within a scenario cluster",
    x = "Scenario cluster",
    y = "Share (%)",
    fill = "County"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank()
  )

print(county_plot)
```

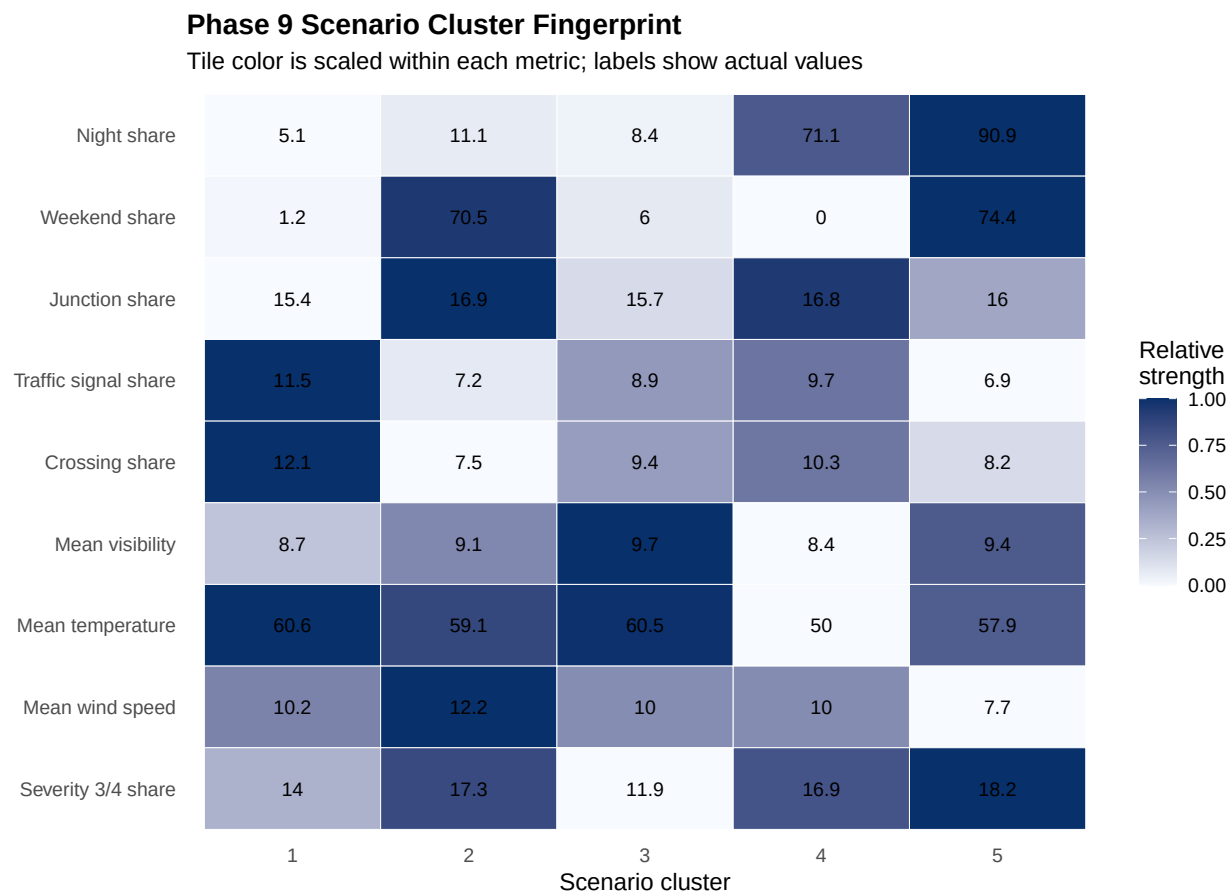



```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_county_composition.png"),
  plot = county_plot,
  width = 7.5,
  height = 4.8,
  dpi = 300
)
```

```r
fingerprint_source <- merge(
  scenario_weather_time_profile[, .(
    scenario_cluster,
    night_pct,
```

```r
    weekend_pct,
    junction_pct,
    traffic_signal_pct,
    crossing_pct,
    mean_visibility,
    mean_temperature,
    mean_wind_speed
  )],
  scenario_severity_profile[, .(scenario_cluster, sev34_pct)],
  by = "scenario_cluster",
  all.x = TRUE
)

fingerprint_long <- data.table::melt(
  fingerprint_source,
  id.vars = "scenario_cluster",
  variable.name = "metric",
  value.name = "metric_value"
)

metric_labels <- c(
  night_pct = "Night share",
  weekend_pct = "Weekend share",
  junction_pct = "Junction share",
  traffic_signal_pct = "Traffic signal share",
  crossing_pct = "Crossing share",
  mean_visibility = "Mean visibility",
  mean_temperature = "Mean temperature",
  mean_wind_speed = "Mean wind speed",
  sev34_pct = "Severity 3/4 share"
)

fingerprint_long[, metric_label := factor(
  metric_labels[metric],
  levels = rev(metric_labels)
)]
fingerprint_long[, scaled_value := rescale01(metric_value), by = metric]

fingerprint_plot <- ggplot2::ggplot(
  fingerprint_long,
  ggplot2::aes(x = factor(scenario_cluster), y = metric_label, fill = scaled_value)
) +
  ggplot2::geom_tile(color = "white") +
  ggplot2::geom_text(ggplot2::aes(label = round(metric_value, 1)), size = 3) +
  ggplot2::scale_fill_gradient(low = "#f7fbff", high = "#08306b") +
  ggplot2::labs(
    title = "Phase 9 Scenario Cluster Fingerprint",
    subtitle = "Tile color is scaled within each metric; labels show actual values",
    x = "Scenario cluster",
    y = NULL,
    fill = "Relative\nstrength"
  ) +
  ggplot2::theme_minimal() +
```

```r
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid = ggplot2::element_blank()
  )

print(fingerprint_plot)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_cluster_fingerprint_heatmap.png"),
  plot = fingerprint_plot,
  width = 8,
  height = 5.8,
  dpi = 300
)
```

```r
write.csv(
  df_clustered,
  file.path(output_dir, "li_accidents_clustered.csv"),
  row.names = FALSE
)

table_exports <- list(
  phase6_validation_indices = validation_table,
  phase6_validation_metadata = validation_metadata,
```

```r
  phase6_stability_summary = stability_summary,
  phase6_selection_summary = selection_summary,
  phase7_eps_trials = eps_trials,
  phase7_parameter_summary = parameter_summary,
  phase7_dbscan_cluster_sizes = spatial_cluster_sizes,
  phase7_corridor_assignments = non_noise_assignments,
  phase8_severity_profile = scenario_severity_profile,
  phase8_county_profile = scenario_county_profile,
  phase8_weather_time_profile = scenario_weather_time_profile,
  phase8_spatial_scenario_crosstab = spatial_scenario_crosstab,
  phase8_spatial_scenario_dominance = spatial_scenario_dominance,
  phase8_scenario_final_names = scenario_final_names,
  phase8_spatial_final_names = spatial_final_names
)

if (!is.null(validation_repeat_results)) {
  table_exports$phase6_validation_repeats <- validation_repeat_results
}

if (!is.null(stability_repeat_results)) {
  table_exports$phase6_stability_repeats <- stability_repeat_results
}

invisible(lapply(names(table_exports), function(table_name) {
  fwrite(table_exports[[table_name]], file.path(table_dir, paste0(table_name, ".csv")))
}))

figure_exports <- c(
  "phase5_kproto_elbow_plot.png",
  "phase6_cluster_profiles.pdf",
  "phase7_knn_distance_plot.png",
  "phase7_spatial_hotspot_map.png",
  "phase7_spatial_density_heatmap.png",
  "phase9_validation_index_plot.png",
  "phase9_stability_plot.png",
  "phase9_spatial_scenario_heatmap.png",
  "phase9_spatial_cluster_facets_heatmap.png",
  "phase9_scenario_risk_bubble.png",
  "phase9_county_composition.png",
  "phase9_cluster_fingerprint_heatmap.png"
)

invisible(lapply(figure_exports, function(file_name) {
  source_path <- file.path(output_dir, file_name)
  if (file.exists(source_path)) {
    file.copy(source_path, file.path(figure_dir, file_name), overwrite = TRUE)
  }
}))

file.copy(
  file.path(output_dir, "li_accidents_clustered.csv"),
  file.path(table_dir, "li_accidents_clustered.csv"),
  overwrite = TRUE
```

```
)
```

```
## [1] TRUE
```

```
export_manifest <- data.table(
  artifact_type = c(
    rep("figure", length(figure_exports)),
    rep("table", length(table_exports)),
    "table"
  ),
  artifact_name = c(
    figure_exports,
    paste0(names(table_exports), ".csv"),
    "li_accidents_clustered.csv"
  ),
  target_path = c(
    file.path(figure_dir, figure_exports),
    file.path(table_dir, paste0(names(table_exports), ".csv")),
    file.path(table_dir, "li_accidents_clustered.csv")
  )
)

fwrite(export_manifest, file.path(output_dir, "phase9_export_manifest.csv"))

knitr::kable(export_manifest, caption = "Phase 9 export manifest")
```

Table 52: Phase 9 export manifest

| artifact_type | artifact_name | target_path |
|---|---|---|
| figure | phase5_kproto_elbow_plot.png | local_artifacts/output/figures/phase5_kproto_elbow_plot.png |
| figure | phase6_cluster_profiles.pdf | local_artifacts/output/figures/phase6_cluster_profiles.pdf |
| figure | phase7_knn_distance_plot.png | local_artifacts/output/figures/phase7_knn_distance_plot.png |
| figure | phase7_spatial_hotspot_map.png | local_artifacts/output/figures/phase7_spatial_hotspot_map.png |
| figure | phase7_spatial_density_heatmap.png | local_artifacts/output/figures/phase7_spatial_density_heatmap.png |
| figure | phase9_validation_index_plot.png | local_artifacts/output/figures/phase9_validation_index_plot.png |
| figure | phase9_stability_plot.png | local_artifacts/output/figures/phase9_stability_plot.png |
| figure | phase9_spatial_scenario_heatmap.png | local_artifacts/output/figures/phase9_spatial_scenario_heatmap.png |
| figure | phase9_spatial_cluster_facets_heatmap.p | local_artifacts/output/figures/phase9_spatial_cluster_facets_heatmap.p |
| figure | phase9_scenario_risk_bubble.png | local_artifacts/output/figures/phase9_scenario_risk_bubble.png |
| figure | phase9_county_composition.png | local_artifacts/output/figures/phase9_county_composition.png |
| figure | phase9_cluster_fingerprint_heatmap.pn | local_artifacts/output/figures/phase9_cluster_fingerprint_heatmap.png |
| table | phase6_validation_indices.csv | local_artifacts/output/tables/phase6_validation_indices.csv |
| table | phase6_validation_metadata.csv | local_artifacts/output/tables/phase6_validation_metadata.csv |
| table | phase6_stability_summary.csv | local_artifacts/output/tables/phase6_stability_summary.csv |
| table | phase6_selection_summary.csv | local_artifacts/output/tables/phase6_selection_summary.csv |
| table | phase7_eps_trials.csv | local_artifacts/output/tables/phase7_eps_trials.csv |
| table | phase7_parameter_summary.csv | local_artifacts/output/tables/phase7_parameter_summary.csv |
| table | phase7_dbscan_cluster_sizes.csv | local_artifacts/output/tables/phase7_dbscan_cluster_sizes.csv |
| table | phase7_corridor_assignments.csv | local_artifacts/output/tables/phase7_corridor_assignments.csv |
| table | phase8_severity_profile.csv | local_artifacts/output/tables/phase8_severity_profile.csv |
| table | phase8_county_profile.csv | local_artifacts/output/tables/phase8_county_profile.csv |

| artifact_type | artifact_name | target_path |
|---|---|---|
| table | phase8_weather_time_profile.csv | local_artifacts/output/tables/phase8_weather_time_profile.csv |
| table | phase8_spatial_scenario_crosstab.csv | local_artifacts/output/tables/phase8_spatial_scenario_crosstab.csv |
| table | phase8_spatial_scenario_dominance.csv | local_artifacts/output/tables/phase8_spatial_scenario_dominance.csv |
| table | phase8_scenario_final_names.csv | local_artifacts/output/tables/phase8_scenario_final_names.csv |
| table | phase8_spatial_final_names.csv | local_artifacts/output/tables/phase8_spatial_final_names.csv |
| table | li_accidents_clustered.csv | local_artifacts/output/tables/li_accidents_clustered.csv |

```
interactive_exports <- data.table(
  artifact_name = c(
    "phase9_validation_index_interactive.html",
    "phase9_stability_interactive.html",
    "phase9_spatial_hotspot_interactive.html",
    "phase9_spatial_density_interactive.html",
    "phase9_scenario_risk_interactive.html",
    "phase9_county_composition_interactive.html",
    "phase9_cluster_fingerprint_interactive.html",
    "phase9_spatial_scenario_heatmap_interactive.html"
  ),
  target_path = file.path(
    interactive_dir,
    c(
      "phase9_validation_index_interactive.html",
      "phase9_stability_interactive.html",
      "phase9_spatial_hotspot_interactive.html",
      "phase9_spatial_density_interactive.html",
      "phase9_scenario_risk_interactive.html",
      "phase9_county_composition_interactive.html",
      "phase9_cluster_fingerprint_interactive.html",
      "phase9_spatial_scenario_heatmap_interactive.html"
    )
  ),
  saved = FALSE,
  note = ifelse(
    interactive_exports_enabled,
    "Saved as interactive HTML",
    "Install plotly and htmlwidgets to enable interactive exports"
  )
)

if (interactive_exports_enabled) {
  spatial_interactive_data <- copy(df_spatial_plot)
  if (nrow(spatial_interactive_data) > 15000) {
    set.seed(42)
    spatial_interactive_data <- spatial_interactive_data[
      ,
      .SD[sample(.N, size = min(.N, max(1L, ceiling(15000 * .N / nrow(df_spatial_plot)))))],
      by = cluster_label
    ]
  }

  spatial_map_interactive <- ggplot2::ggplot(
```

```
    spatial_interactive_data,
    ggplot2::aes(x = x_km, y = y_km, color = cluster_label,
                 text = paste0(
                   "Cluster: ", cluster_label,
                   "<br>x_km: ", round(x_km, 2),
                   "<br>y_km: ", round(y_km, 2)
                 ))
  ) +
    ggplot2::geom_point(size = 0.6, alpha = 0.7) +
    ggplot2::scale_color_manual(values = cluster_colors) +
    ggplot2::labs(
      title = "Interactive Spatial Hotspot View",
      subtitle = "Rendered on a representative sample for smoother browsing",
      x = "Projected x (km)",
      y = "Projected y (km)",
      color = "Cluster"
    ) +
    ggplot2::theme_minimal()

  interactive_widgets <- list(
    phase9_validation_index_interactive.html = plotly::ggplotly(validation_plot, tooltip = c("x", "y"))
    phase9_stability_interactive.html = plotly::ggplotly(stability_plot, tooltip = c("x", "y", "fill"))
    phase9_spatial_hotspot_interactive.html = plotly::ggplotly(spatial_map_interactive, tooltip = "text
    phase9_spatial_density_interactive.html = plotly::plot_ly(
      spatial_interactive_data,
      x = ~x_km,
      y = ~y_km,
      type = "histogram2d",
      colorscale = "Viridis",
      nbinsx = 80,
      nbinsy = 60
    ) |>
      plotly::layout(
        title = "Interactive Spatial Density Heatmap",
        xaxis = list(title = "Projected x (km)"),
        yaxis = list(title = "Projected y (km)")
      ),
    phase9_scenario_risk_interactive.html = plotly::ggplotly(risk_bubble_plot, tooltip = c("x", "y", "s
    phase9_county_composition_interactive.html = plotly::ggplotly(county_plot, tooltip = c("x", "y", "f
    phase9_cluster_fingerprint_interactive.html = plotly::ggplotly(fingerprint_plot, tooltip = c("x", "
    phase9_spatial_scenario_heatmap_interactive.html = plotly::ggplotly(heatmap_plot, tooltip = c("x",
  )

  invisible(lapply(names(interactive_widgets), function(file_name) {
    htmlwidgets::saveWidget(
      interactive_widgets[[file_name]],
      file = file.path(interactive_dir, file_name),
      selfcontained = FALSE
    )
  }))

  interactive_exports[, saved := file.exists(target_path)]
}
```

```r
fwrite(interactive_exports, file.path(output_dir, "phase9_interactive_manifest.csv"))

knitr::kable(interactive_exports, caption = "Phase 9 interactive export manifest")
```

Table 53: Phase 9 interactive export manifest

| artifact_name | target_path | saved | note |
|---|---|---|---|
| phase9_validation_index_interactive.html | local_artifacts/output/interactive/phase9_validation_index_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |
| phase9_stability_interactive.html | local_artifacts/output/interactive/phase9_stability_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |
| phase9_spatial_hotspot_interactive.html | local_artifacts/output/interactive/phase9_spatial_hotspot_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |
| phase9_spatial_density_interactive.html | local_artifacts/output/interactive/phase9_spatial_density_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |
| phase9_scenario_risk_interactive.html | local_artifacts/output/interactive/phase9_scenario_risk_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |
| phase9_county_composition_interactive.html | local_artifacts/output/interactive/phase9_county_composition_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |
| phase9_cluster_fingerprint_interactive.html | local_artifacts/output/interactive/phase9_cluster_fingerprint_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |
| phase9_spatial_scenario_heatmap_interactive.html | local_artifacts/output/interactive/phase9_spatial_scenario_heatmap_interactive.html | FALSE | Install plotly and htmlwidgets to enable interactive exports |

```r
phase_9_summary <- data.table(
  checkpoint = c(
    "Figure directory created",
    "Table directory created",
    "Validation plot saved",
    "Stability plot saved",
    "Heatmap saved",
    "Spatial density heatmap saved",
    "Spatial cluster facet heatmap saved",
    "Scenario risk bubble saved",
    "County composition plot saved",
    "Cluster fingerprint heatmap saved",
    "Labeled CSV exported",
    "Interactive manifest saved",
    "Export manifest saved"
  ),
  status = c(
    dir.exists(figure_dir),
    dir.exists(table_dir),
    file.exists(file.path(figure_dir, "phase9_validation_index_plot.png")),
    file.exists(file.path(figure_dir, "phase9_stability_plot.png")),
    file.exists(file.path(figure_dir, "phase9_spatial_scenario_heatmap.png")),
    file.exists(file.path(figure_dir, "phase7_spatial_density_heatmap.png")),
    file.exists(file.path(figure_dir, "phase9_spatial_cluster_facets_heatmap.png")),
    file.exists(file.path(figure_dir, "phase9_scenario_risk_bubble.png")),
    file.exists(file.path(figure_dir, "phase9_county_composition.png")),
    file.exists(file.path(figure_dir, "phase9_cluster_fingerprint_heatmap.png")),
```

```r
    file.exists(file.path(table_dir, "li_accidents_clustered.csv")),
    file.exists(file.path(output_dir, "phase9_interactive_manifest.csv")),
    file.exists(file.path(output_dir, "phase9_export_manifest.csv"))
  )
)

fwrite(phase_9_summary, file.path(output_dir, "phase9_summary.csv"))

knitr::kable(phase_9_summary, caption = "Phase 9 readiness summary")
```

Table 54: Phase 9 readiness summary

| checkpoint | status |
|---|---|
| Figure directory created | TRUE |
| Table directory created | TRUE |
| Validation plot saved | TRUE |
| Stability plot saved | TRUE |
| Heatmap saved | TRUE |
| Spatial density heatmap saved | TRUE |
| Spatial cluster facet heatmap saved | TRUE |
| Scenario risk bubble saved | TRUE |
| County composition plot saved | TRUE |
| Cluster fingerprint heatmap saved | TRUE |
| Labeled CSV exported | TRUE |
| Interactive manifest saved | TRUE |
| Export manifest saved | TRUE |

## Phase 10: Final Report and Submission Checklist

This phase keeps a simple readiness tracker for the final report assets and submission handoff.

```r
phase_10_checklist <- data.table(
  deliverable = c(
    "Long Island SRS source",
    "Execution plan source",
    "Main workflow Rmd",
    "Cleaned dataset",
    "Clustered labeled dataset",
    "Figure export folder",
    "Table export folder",
    "Interactive export manifest"
  ),
  path = c(
    "Li_Clustering_SRS.tex",
    "Li_Clustering_Execution_Plan.tex",
    "li_clustering.Rmd",
    input_path,
    file.path(table_dir, "li_accidents_clustered.csv"),
    figure_dir,
    table_dir,
```

```r
    file.path(output_dir, "phase9_interactive_manifest.csv")
  ),
  ready = c(
    file.exists("Li_Clustering_SRS.tex"),
    file.exists("Li_Clustering_Execution_Plan.tex"),
    file.exists("li_clustering.Rmd"),
    file.exists(input_path),
    file.exists(file.path(table_dir, "li_accidents_clustered.csv")),
    dir.exists(figure_dir),
    dir.exists(table_dir),
    file.exists(file.path(output_dir, "phase9_interactive_manifest.csv"))
  )
)

fwrite(phase_10_checklist, file.path(output_dir, "phase10_readiness_checklist.csv"))

knitr::kable(phase_10_checklist, caption = "Phase 10 readiness checklist")
```

Table 55: Phase 10 readiness checklist

| deliverable | path | ready |
|---|---|---|
| Long Island SRS source | Li_Clustering_SRS.tex | TRUE |
| Execution plan source | Li_Clustering_Execution_Plan.tex | TRUE |
| Main workflow Rmd | li_clustering.Rmd | TRUE |
| Cleaned dataset | ../li_accidents_clean.csv | TRUE |
| Clustered labeled dataset | local_artifacts/output/tables/li_accidents_clustered.csv | TRUE |
| Figure export folder | local_artifacts/output/figures | TRUE |
| Table export folder | local_artifacts/output/tables | TRUE |
| Interactive export manifest | local_artifacts/output/phase9_interactive_manifest.csv | TRUE |

```r
phase_10_report_outline <- data.table(
  report_section = c(
    "Introduction and motivation",
    "Dataset description and audit findings",
    "Feature engineering decisions",
    "Method selection rationale",
    "Scenario clustering results",
    "Spatial hotspot results",
    "Interpretation and cross-tab findings",
    "Limitations",
    "Conclusion and safety priorities"
  ),
  main_source = c(
    "Li_Clustering_SRS.tex",
    file.path(table_dir, "phase8_severity_profile.csv"),
    file.path(table_dir, "phase8_weather_time_profile.csv"),
    "Li_Clustering_Execution_Plan.tex",
    file.path(figure_dir, "phase9_validation_index_plot.png"),
    file.path(figure_dir, "phase7_spatial_hotspot_map.png"),
    file.path(table_dir, "phase8_spatial_scenario_crosstab.csv"),
    file.path(output_dir, "phase10_limitations_table.csv"),
    file.path(output_dir, "phase10_research_question_coverage.csv")
```

```
  ),
  ready = c(
    file.exists("Li_Clustering_SRS.tex"),
    file.exists(file.path(table_dir, "phase8_severity_profile.csv")),
    file.exists(file.path(table_dir, "phase8_weather_time_profile.csv")),
    file.exists("Li_Clustering_Execution_Plan.tex"),
    file.exists(file.path(figure_dir, "phase9_validation_index_plot.png")),
    file.exists(file.path(figure_dir, "phase7_spatial_hotspot_map.png")),
    file.exists(file.path(table_dir, "phase8_spatial_scenario_crosstab.csv")),
    TRUE,
    TRUE
  )
)

fwrite(phase_10_report_outline, file.path(output_dir, "phase10_report_outline.csv"))

knitr::kable(phase_10_report_outline, caption = "Phase 10 report outline and source map")
```

Table 56: Phase 10 report outline and source map

| report_section | main_source | ready |
|---|---|---|
| Introduction and motivation | Li_Clustering_SRS.tex | TRUE |
| Dataset description and audit findings | local_artifacts/output/tables/phase8_severity_profile.csv | TRUE |
| Feature engineering decisions | local_artifacts/output/tables/phase8_weather_time_profile.csv | TRUE |
| Method selection rationale | Li_Clustering_Execution_Plan.tex | TRUE |
| Scenario clustering results | local_artifacts/output/figures/phase9_validation_index_plot.png | TRUE |
| Spatial hotspot results | local_artifacts/output/figures/phase7_spatial_hotspot_map.png | TRUE |
| Interpretation and cross-tab findings | local_artifacts/output/tables/phase8_spatial_scenario_crosstab.csv | TRUE |
| Limitations | local_artifacts/output/phase10_limitations_table.csv | TRUE |
| Conclusion and safety priorities | local_artifacts/output/phase10_research_question_coverage.csv | TRUE |

```
phase_10_limitations <- data.table(
  limitation = c(
    "Duration capping",
    "Precipitation exclusion",
    "Distance exclusion",
    "Severity 2 dominance",
    "Sampled Phase 6 validation",
    "DBSCAN parameter sensitivity"
  ),
  report_language = c(
    "Duration was capped at the 99th percentile to reduce the effect of extreme outliers, so the top 1%
    "Precipitation was excluded from the core model because missingness was high enough to make routine
    "Distance(mi) was excluded because the large share of zero values suggests a reporting artifact rat
    "Severity 2 dominates the dataset, so subtle differences among severe crashes can be harder to sepa
    if (run_profile == "full_srs") {
      "Phase 6 is configured for full-data validation and stability on a stronger machine, so this sett
    } else if (run_profile == "consensus_16gb") {
      paste(
        "Phase 6 uses repeated sampled validation and repeated sampled stability checks",
```

```
        "to build a consensus choice of k on a 16 GB machine;",
        "the archived full-data version is still available for a later high-memory rerun."
      )
    } else {
      "Phase 6 uses sampled silhouette validation and sampled stability checks on this machine to stay w
    },
    "Spatial hotspot shapes can shift if eps or minPts changes, so corridor labels should be read as dat
  ),
  include_in_final_report = TRUE
)

fwrite(phase_10_limitations, file.path(output_dir, "phase10_limitations_table.csv"))

knitr::kable(phase_10_limitations, caption = "Phase 10 limitations table")
```

Table 57: Phase 10 limitations table

| limitation | report_language | include_in_final_report |
|---|---|---|
| Duration capping | Duration was capped at the 99th percentile to reduce the effect of extreme outliers, so the top 1% of incidents lose some real variation. | TRUE |
| Precipitation exclusion | Precipitation was excluded from the core model because missingness was high enough to make routine imputation less defensible. | TRUE |
| Distance exclusion | Distance(mi) was excluded because the large share of zero values suggests a reporting artifact rather than a stable behavioral feature. | TRUE |
| Severity 2 dominance | Severity 2 dominates the dataset, so subtle differences among severe crashes can be harder to separate. | TRUE |
| Sampled Phase 6 validation | Phase 6 uses sampled silhouette validation and sampled stability checks on this machine to stay within local memory limits; the full-data version is archived for a later high-memory rerun. | TRUE |
| DBSCAN parameter sensitivity | Spatial hotspot shapes can shift if eps or minPts changes, so corridor labels should be read as data-driven approximations. | TRUE |

```
phase_10_rq_coverage <- data.table(
  research_focus = c(
    "Scenario accident archetypes",
    "Spatial hotspot corridors",
    "High-risk severity patterns",
    "County composition differences",
    "Hotspot-to-scenario linkage"
  ),
  evidence_source = c(
    file.path(table_dir, "phase8_scenario_final_names.csv"),
    file.path(table_dir, "phase8_spatial_final_names.csv"),
    file.path(table_dir, "phase8_severity_profile.csv"),
    file.path(table_dir, "phase8_county_profile.csv"),
    file.path(table_dir, "phase8_spatial_scenario_crosstab.csv")
  ),
  ready = c(
    file.exists(file.path(table_dir, "phase8_scenario_final_names.csv")),
    file.exists(file.path(table_dir, "phase8_spatial_final_names.csv")),
    file.exists(file.path(table_dir, "phase8_severity_profile.csv")),
```

```r
    file.exists(file.path(table_dir, "phase8_county_profile.csv")),
    file.exists(file.path(table_dir, "phase8_spatial_scenario_crosstab.csv"))
  )
)

fwrite(phase_10_rq_coverage, file.path(output_dir, "phase10_research_question_coverage.csv"))

knitr::kable(phase_10_rq_coverage, caption = "Phase 10 research-question coverage")
```

Table 58: Phase 10 research-question coverage

| research_focus | evidence_source | ready |
|---|---|---|
| Scenario accident archetypes | local_artifacts/output/tables/phase8_scenario_final_names.csv | TRUE |
| Spatial hotspot corridors | local_artifacts/output/tables/phase8_spatial_final_names.csv | TRUE |
| High-risk severity patterns | local_artifacts/output/tables/phase8_severity_profile.csv | TRUE |
| County composition differences | local_artifacts/output/tables/phase8_county_profile.csv | TRUE |
| Hotspot-to-scenario linkage | local_artifacts/output/tables/phase8_spatial_scenario_crosstab.csv | TRUE |

```r
phase_10_submission_manifest <- data.table(
  deliverable_group = c(
    "Core source",
    "Core source",
    "Core source",
    "Data",
    "Data",
    "Exports",
    "Exports",
    "Exports"
  ),
  deliverable = c(
    "Li_Clustering_SRS.tex",
    "Li_Clustering_Execution_Plan.tex",
    "li_clustering.Rmd",
    "li_accidents_clean.csv",
    "li_accidents_clustered.csv",
    "Figure directory",
    "Table directory",
    "Phase 10 checklist bundle"
  ),
  path = c(
    "Li_Clustering_SRS.tex",
    "Li_Clustering_Execution_Plan.tex",
    "li_clustering.Rmd",
    input_path,
    file.path(table_dir, "li_accidents_clustered.csv"),
    figure_dir,
    table_dir,
    output_dir
  ),
  ready = c(
    file.exists("Li_Clustering_SRS.tex"),
```

```r
    file.exists("Li_Clustering_Execution_Plan.tex"),
    file.exists("li_clustering.Rmd"),
    file.exists(input_path),
    file.exists(file.path(table_dir, "li_accidents_clustered.csv")),
    dir.exists(figure_dir),
    dir.exists(table_dir),
    dir.exists(output_dir)
  )
)

fwrite(phase_10_submission_manifest, file.path(output_dir, "phase10_submission_manifest.csv"))

knitr::kable(phase_10_submission_manifest, caption = "Phase 10 submission manifest")
```

Table 59: Phase 10 submission manifest

| deliverable_group | deliverable | path | ready |
|---|---|---|---|
| Core source | Li_Clustering_SRS.tex | Li_Clustering_SRS.tex | TRUE |
| Core source | Li_Clustering_Execution_Plan.tex | Li_Clustering_Execution_Plan.tex | TRUE |
| Core source | li_clustering.Rmd | li_clustering.Rmd | TRUE |
| Data | li_accidents_clean.csv | ../li_accidents_clean.csv | TRUE |
| Data | li_accidents_clustered.csv | local_artifacts/output/tables/li_accidents_clustered.csv | TRUE |
| Exports | Figure directory | local_artifacts/output/figures | TRUE |
| Exports | Table directory | local_artifacts/output/tables | TRUE |
| Exports | Phase 10 checklist bundle | local_artifacts/output | TRUE |

```r
phase_10_summary <- data.table(
  checkpoint = c(
    "Readiness checklist saved",
    "Report outline saved",
    "Limitations table saved",
    "Research-question coverage saved",
    "Submission manifest saved"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase10_readiness_checklist.csv")),
    file.exists(file.path(output_dir, "phase10_report_outline.csv")),
    file.exists(file.path(output_dir, "phase10_limitations_table.csv")),
    file.exists(file.path(output_dir, "phase10_research_question_coverage.csv")),
    file.exists(file.path(output_dir, "phase10_submission_manifest.csv"))
  )
)

fwrite(phase_10_summary, file.path(output_dir, "phase10_summary.csv"))

knitr::kable(phase_10_summary, caption = "Phase 10 readiness summary")
```

Table 60: Phase 10 readiness summary

| checkpoint | status |
|---|---|
| Readiness checklist saved | TRUE |
| Report outline saved | TRUE |
| Limitations table saved | TRUE |
| Research-question coverage saved | TRUE |
| Submission manifest saved | TRUE |

```r
selected_jaccard_value <- suppressWarnings(as.numeric(selected_jaccard))

phase_10_sanity_audit <- data.table(
  check = c(
    "Scenario names are unique",
    "Scenario labels have no gaps",
    "Spatial labels have no gaps",
    "Selected scenario stability is at least moderate",
    "Largest spatial cluster stays below 60% of all rows",
    "DBSCAN cluster-count target met",
    "DBSCAN noise target met"
  ),
  status = c(
    uniqueN(scenario_final_names$final_name) == nrow(scenario_final_names),
    label_join_check[metric == "Scenario label NAs", value][[1]] == 0,
    label_join_check[metric == "Spatial label NAs", value][[1]] == 0,
    !is.na(selected_jaccard_value) && selected_jaccard_value >= 0.4,
    largest_non_noise_share_pct <= phase7_max_cluster_share_cap,
    uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) >= 3 &
      uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) <= 15,
    round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) >= 5 &
      round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) <= 30
  ),
  note = c(
    "Final scenario names should be distinct for presentation and discussion.",
    "Every row should carry a scenario label after the join.",
    "Every row should carry a spatial label after the join.",
    "A Jaccard score below 0.4 should be presented as exploratory rather than settled.",
    "This avoids one hotspot absorbing most of Long Island.",
    "The SRS target is between 3 and 15 non-noise clusters.",
    "The SRS target is between 5% and 30% noise."
  )
)

fwrite(phase_10_sanity_audit, file.path(output_dir, "phase10_sanity_audit.csv"))

phase_10_summary <- rbind(
  phase_10_summary,
  data.table(
    checkpoint = "Sanity audit saved",
    status = file.exists(file.path(output_dir, "phase10_sanity_audit.csv"))
  )
)
fwrite(phase_10_summary, file.path(output_dir, "phase10_summary.csv"))
```

```r
knitr::kable(phase_10_sanity_audit, caption = "Phase 10 sanity audit")
```

Table 61: Phase 10 sanity audit

| check | status | note |
|---|---|---|
| Scenario names are unique | TRUE | Final scenario names should be distinct for presentation and discussion. |
| Scenario labels have no gaps | TRUE | Every row should carry a scenario label after the join. |
| Spatial labels have no gaps | TRUE | Every row should carry a spatial label after the join. |
| Selected scenario stability is at least moderate | FALSE | A Jaccard score below 0.4 should be presented as exploratory rather than settled. |
| Largest spatial cluster stays below 60% of all rows | TRUE | This avoids one hotspot absorbing most of Long Island. |
| DBSCAN cluster-count target met | TRUE | The SRS target is between 3 and 15 non-noise clusters. |
| DBSCAN noise target met | TRUE | The SRS target is between 5% and 30% noise. |

## Notes for the next phases

- Phase 2 should stay read-only and focus on profiling the cleaned dataset.
- Phase 3 now derives the main engineered features needed for the scenario and spatial matrices.
- Phase 4 now defines the actual scenario and spatial matrices for clustering.
- Phase 5 now sets up the candidate k-prototypes runs and their core diagnostics.
- Phase 6 now validates the candidate solutions, checks stability, and assigns a selected k.
- Phase 7 now sets DBSCAN parameters, creates the spatial hotspot solution, and assigns corridor-style labels.
- Phase 8 now joins cluster labels back to the working data and builds the interpretation tables.
- Phase 9 now exports the report-ready figures, tables, and labeled dataset.
- Phase 10 now tracks final report and submission readiness.
- `run_profile = "consensus_16gb"` is the current default in this file.
- `run_profile = "local_safe"` is the quickest laptop-friendly option.
- `run_profile = "consensus_16gb"` repeats sampled Phase 6 checks and aggregates them into a consensus choice for 16 GB machines.
- `run_profile = "full_srs"` is reserved for a stronger machine that can support the full Phase 6 diagnostics.
- The current default uses repeated sampled validation and repeated sampled stability checks to stay within 16 GB memory limits.
- Interactive HTML exports are optional and now turn on automatically when the needed packages are available.
- The original full-data Phase 6 validation and stability calls are archived in `eval=FALSE` chunks for later reruns.
- Phase 3 and later should write tables, figures, and exports into `local_artifacts/output/`.
- No `.RData` workspace files are needed for this workflow.

```r
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4
##
```

```
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versi
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] rmarkdown_2.30      knitr_1.51          factoextra_2.0.0
##  [4] geosphere_1.6-8     ggplot2_4.0.2       fpc_2.2-14
##  [7] cluster_2.1.8.2     dbscan_1.2.4        clustMixType_0.4-2
## [10] lubridate_1.9.5     dplyr_1.2.0         data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4     class_7.3-23       robustbase_0.99-7  lattice_0.22-7
##  [5] digest_0.6.39      magrittr_2.0.4     evaluate_1.0.5     grid_4.5.2
##  [9] timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0      ggrepel_0.9.8
## [13] nnet_7.3-20        mclust_6.1.2       combinat_0.0-8     kernlab_0.9-33
## [17] viridisLite_0.4.3  scales_1.4.0       modeltools_0.2-24  cli_3.6.5
## [21] rlang_1.1.7        withr_3.0.2        yaml_2.3.12        otel_0.2.0
## [25] tools_4.5.2        flexmix_2.3-20     parallel_4.5.2     vctrs_0.7.1
## [29] R6_2.6.1           stats4_4.5.2       lifecycle_1.0.5    htmlwidgets_1.6.4
## [33] MASS_7.3-65        pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6
## [37] glue_1.8.0         Rcpp_1.1.1         DEoptimR_1.1-4     xfun_0.56
## [41] tibble_3.3.1       tidyselect_1.2.1   prabclus_2.3-5     farver_2.1.2
## [45] htmltools_0.5.9    labeling_0.4.3     compiler_4.5.2     S7_0.2.1
## [49] diptest_0.77-2
```